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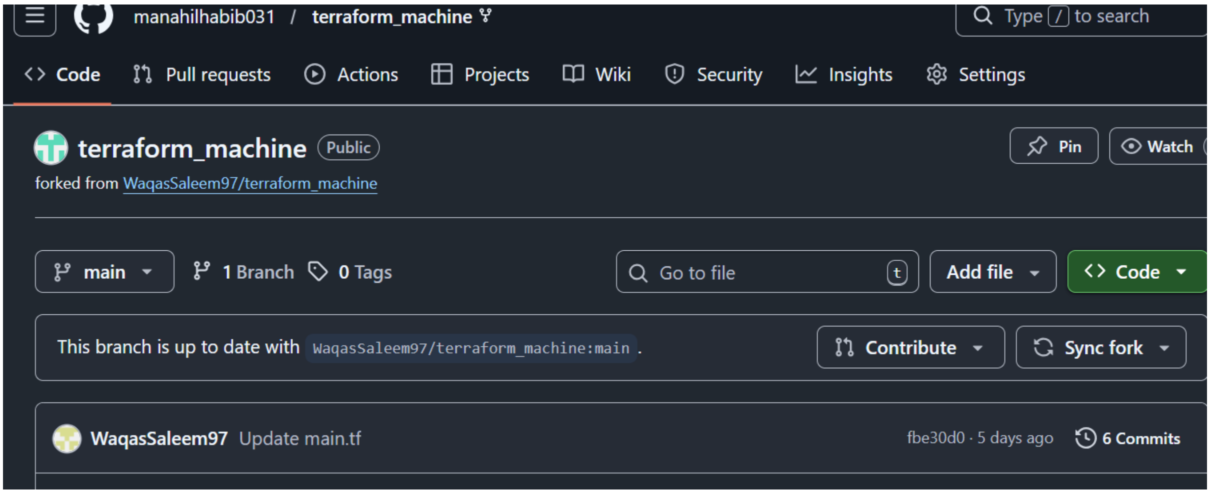
Reg no: 2023-BSE-042

Section: 5B

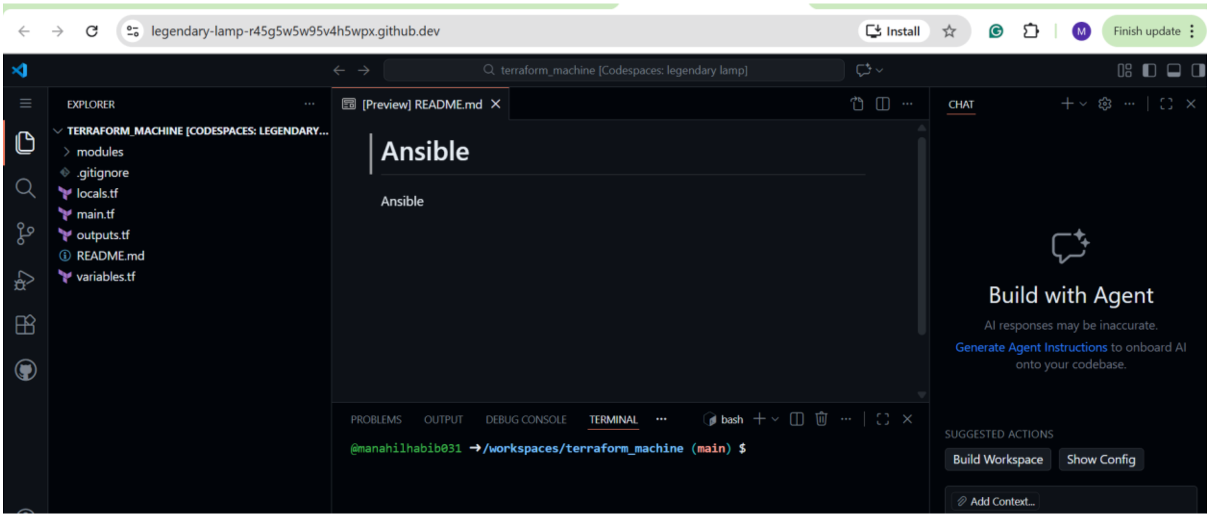
Lab 14

Task 0 – Lab Setup (Codespace & GH CLI)

1. Fork the repo [terraform_machine]



2. **Create/open Codespace** on your GitHub account (from terraform_machine repo).



3. Inside the Codespace terminal, verify:

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ aws --version
aws-cli/2.32.34 Python/3.13.11 Linux/6.8.0-1030-azure exe/x86_64.ubuntu.24
@manahilhabib031 →/workspaces/terraform_machine (main) $ terraform --version
Terraform v1.14.3
on linux_amd64
@manahilhabib031 →/workspaces/terraform_machine (main) $ ansible --version || ec
ho "ansible not yet installed"
bash: ansible: command not found
ansible not yet installed
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

3. Ensure AWS CLI is configured with credentials that have permissions to create EC2, VPC, subnets, and security groups in region **me-central-1**:

aws sts get-caller-identity

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ aws sts get-caller-iden
tity
{
    "UserId": "AIDARMIC7HAZDK6HLORCS",
    "Account": "095032391730",
    "Arn": "arn:aws:iam::095032391730:user/labs"
}
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

Task 1 – Generate ssh key and Initial Terraform apply

You will start from an existing repository and prepare Terraform variables and SSH keys.

1. **Check SSH directory & generate SSH key pair** if not already present:

ls ~/.ssh

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ls ~/.ssh
ls: cannot access '/home/codespace/.ssh': No such file or directory
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

ssh-keygen -t ed25519 -f ~/.ssh/id_ed25519 -N ""

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ssh-keygen -t ed25519 -
f ~/.ssh/id_ed25519 -N ""
Your public key has been saved in /home/codespace/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:lg0vfTAQU7NVh3pCJ69cwMVH4Xqs00b2aFkzY6HbA+4 codespace@codespaces-703057
The key's randomart image is:
+--[ED25519 256]--+
|        +oo..+o++|
|         o += =..|
|        . +. * + |
|         * oo B .|
|        S +..O X.|
|       . . .+ @+=|
|             =+=.|
|            ..o .|
|             E   |
+----[SHA256]-----+
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

ls -la ~/.ssh

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ls -la ~/.ssh
total 20
drwx------ 2 codespace codespace 4096 Jan 14 09:13 .
drwxr-x--- 1 codespace codespace 4096 Jan 14 09:13 ..
-rw------- 1 codespace codespace  419 Jan 14 09:13 id_ed25519
-rw-r--r-- 1 codespace codespace  109 Jan 14 09:13 id_ed25519.pub
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

2. **Create terraform.tfvars** in the repo root:

cd /workspaces/terraform_machine

touch terraform.tfvars

ls -la terraform.tfvars

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cd /workspaces/terraform_machine
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ touch terraform.tfvars
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ls -la terraform.tfvars
-rw-rw-rw- 1 codespace codespace 0 Jan 14 09:15 terraform.tfvars
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

Add the following content:

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat terraform.tfvars
vpc_cidr_block = "10.0.0.0/16"
subnet_cidr_block = "10.0.10.0/24"
availability_zone = "me-central-1a"
env_prefix = "dev"
instance_type = "t3.micro"
public_key = "~/.ssh/id_ed25519.pub"
private_key = "~/.ssh/id_ed25519"
```

3. **Initialize Terraform**:

terraform init

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform init
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

4. **Apply Terraform** to create **2 EC2 instances** (as defined in the existing Terraform code):

terraform apply -auto-approve

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform apply -auto-approve
module.myapp-webserver[1].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[1].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[1].aws_instance.myapp-server: Creation complete after 14s [id=i-00e0
ab7ba73f09261]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 14s [id=i-0273
ccfb211e43a86]

Apply complete! Resources: 10 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "3.28.134.239",
  "3.28.136.139",
]
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

5. **Check outputs**:

terraform output

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.134.239",
  "3.28.136.139",
]
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Task 2 – Static Ansible inventory with two EC2 instances

You will install Ansible (via pipx), create a static inventory, and verify connectivity.

1. **Install Ansible (core) using pipx**:

pipx install ansible-core

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ pipx install ansible-core
  installed package ansible-core 2.20.1, installed using Python 3.12.1
  These apps are now globally available
    - ansible
    - ansible-config
    - ansible-console
    - ansible-doc
    - ansible-galaxy
    - ansible-inventory
    - ansible-playbook
    - ansible-pull
    - ansible-test
    - ansible-vault
done! ✨ 🌟 ✨
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible --version
ansible [core 2.20.1]
  config file = None
  configured module search path = ['/home/codespace/.ansible/plugins/modules', '/usr/share/
ansible/plugins/modules']
  ansible python module location = /usr/local/py-utils/venvs/ansible-core/lib/python3.12/si
te-packages/ansible
  ansible collection location = /home/codespace/.ansible/collections:/usr/share/ansible/col
lections
  executable location = /usr/local/py-utils/bin/ansible
  python version = 3.12.1 (main, Nov 27 2025, 10:47:52) [GCC 13.3.0] (/usr/local/py-utils/v
envs/ansible-core/bin/python)
  jinja version = 3.1.6
  pyyaml version = 6.0.3 (with libyaml v0.2.5)
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Obtain the two public IPs of your EC2 instances:

terraform output

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.134.239",
  "3.28.136.139",
]
```

3. **Create Ansible inventory file hosts**:

touch hosts

ls -la hosts

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ touch hosts
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ls -la hosts
-rw-rw-rw- 1 codespace codespace 0 Jan 14 09:26 hosts
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Add the following (**replace <public-ip-ec2> with your 2 real IPs**):

<public-ip-ec2> ansible_user=ec2-user   ansible_ssh_private_key_file=~/.ssh/id_ed25519

<public-ip-ec2> ansible_user=ec2-user   ansible_ssh_private_key_file=~/.ssh/id_ed25519

```
 GNU nano 7.2                              hosts *
3.28.134.239 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
3.28.136.139 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
```

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ nano hosts
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ cat hosts
3.28.134.239 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
3.28.136.139 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

4. **Test connectivity**:

ansible all -i hosts -m ping

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
Origin: <adhoc 'ping' task>

{'action': 'ping', 'args': {}, 'timeout': 0, 'async_val': 0, 'poll': 15}

3.28.134.239 | UNREACHABLE! => {
    "changed": false,
    "msg": "Task failed: Failed to connect to the host via ssh: Host key verification faile
d.",
    "unreachable": true
}
3.28.136.139 | UNREACHABLE! => {
    "changed": false,
    "msg": "Task failed: Failed to connect to the host via ssh: Host key verification faile
d.",
    "unreachable": true
}
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

5. If it fails due to host key checking, **add** this to each line in hosts:

ansible_ssh_common_args='-o StrictHostKeyChecking=no'

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ nano hosts
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ cat hosts
3.28.134.239 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_s
sh_common_args='-o StrictHostKeyChecking=no'
3.28.136.139 ansible_user=ec2-user ansible_ssh_private_key_file=~/.ssh/id_ed25519 ansible_s
sh_common_args='-o StrictHostKeyChecking=no'
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

6. Retry:

ansible all -i hosts -m ping

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.136.139' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.136.139 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Task 3 - Scale to three instances & group-based inventory

You will expand to 3 web servers via Terraform’s count and restructure your inventory into groups.

1. **Update your Terraform module for webservers** in main.tf to use count = 3:

```
  GNU nano 7.2                          main.tf *
module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id

  # Loop count
  count           = 3
  # Use count.index to differentiate instances
  instance_suffix = count.index

^G Help       ^O Write Out   ^W Where Is   ^K Cut     ^T Execute   ^C L
^X Exit       ^R Read File   ^\ Replace    ^U Paste   ^J Justify   ^/ G
```

Apply Terraform to get 3 instances:

terraform apply -auto-approve

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ terraform apply -auto-approve
Changes to Outputs:
  ~ webserver_public_ips = [
        # (1 unchanged element hidden)
        "3.28.136.139",
      + (known after apply),
    ]
module.myapp-webserver[2].aws_key_pair.ssh-key: Creating...
module.myapp-webserver[2].aws_security_group.web_sg: Creating...
module.myapp-webserver[2].aws_key_pair.ssh-key: Creation complete after 0s [id=dev-serverke
y-2]
module.myapp-webserver[2].aws_security_group.web_sg: Creation complete after 3s [id=sg-00a1
e130ab16b5098]
module.myapp-webserver[2].aws_instance.myapp-server: Creating...
module.myapp-webserver[2].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[2].aws_instance.myapp-server: Creation complete after 13s [id=i-0653
dafbf03021d5b]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "3.28.134.239",
  "3.28.136.139",
  "3.29.129.208",
]
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

3. **Check outputs**:

terraform output

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.134.239",
  "3.28.136.139",
  "3.29.129.208",
]
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

4. **Rewrite your hosts file** using group definitions:

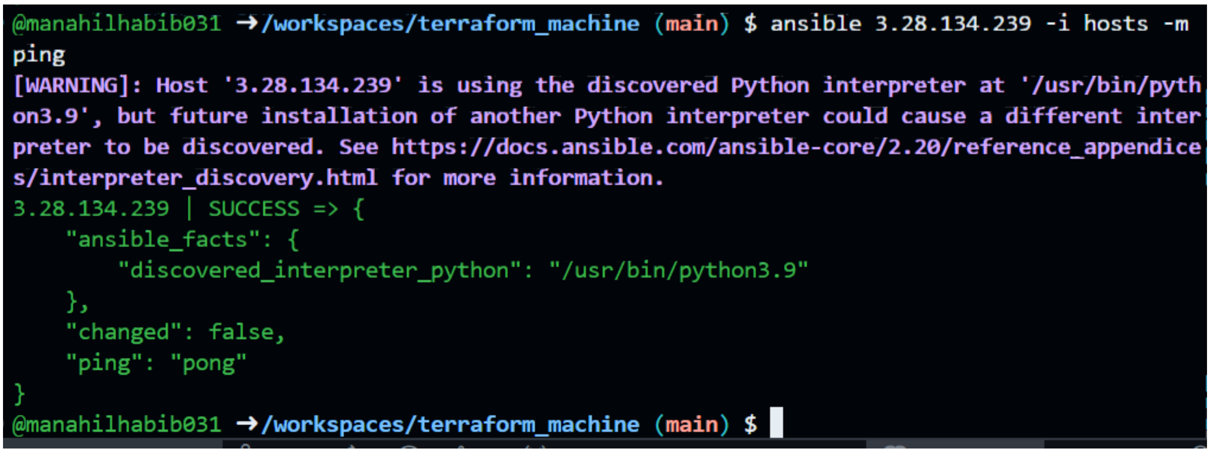
```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat hosts

[ec2]
3.28.134.239
3.28.136.139

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[droplet]
3.29.129.208

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

5. **Test group connectivity**:

ansible ec2 -i hosts -m ping

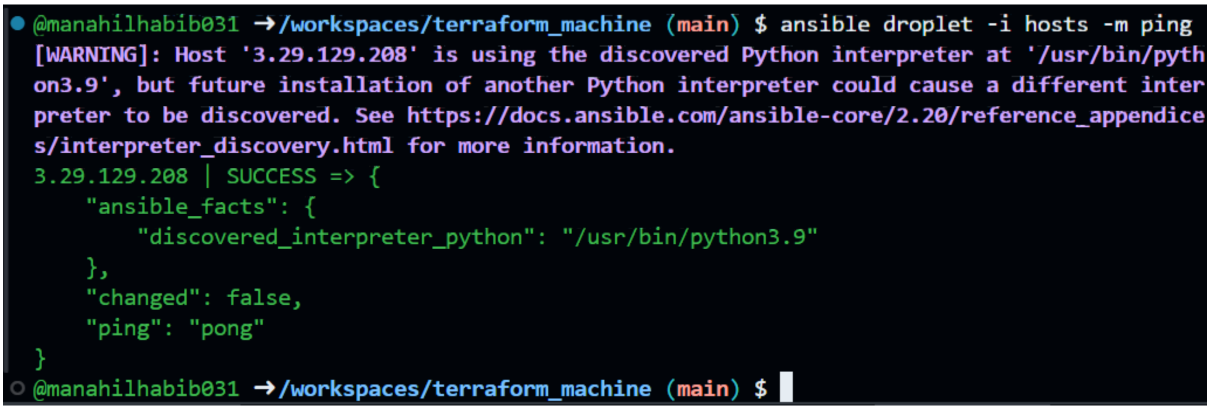
```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ansible ec2 -i hosts -m ping
[WARNING]: Host '3.28.134.239' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.134.239 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.136.139' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.136.139 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

6. **Test single host by IP**:



7. **Test droplet group**:

ansible droplet -i hosts -m ping



8. **Test all hosts**:

ansible all -i hosts -m ping

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.134.239' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.134.239 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.136.139' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.136.139 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Task 4 – Global ansible.cfg & first nginx playbook

You will configure a **global Ansible configuration file**, then create a basic playbook for nginx.

1. **Create global Ansible configuration**:

vim ~/.ansible.cfg

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ vim ~/.ansible.cfg
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ cat ~/.ansible.cfg
[default]
host_key_checking = False
interpreter_python = /usr/bin/python3

@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

2. **Remove ansible_ssh_common_args** from hosts (delete from all groups).

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ cat hosts

[ec2]
3.28.134.239
3.28.136.139

[ec2:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[droplet]
3.29.129.208

[droplet:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/id_ed25519

@manahilhabib031 →/workspaces/terraform_machine (main) $
```

3. **Confirm connectivity**:

ansible all -i hosts -m ping

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ansible all -i hosts -m ping
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.136.139' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.136.139 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host '3.28.134.239' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
3.28.134.239 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

4. **Create my-playbook.yaml**:

touch my-playbook.yaml

ls -la my-playbook.yaml

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ touch my-playbook.yaml
@manahilhabib031 →/workspaces/terraform_machine (main) $ ls -la my-playbook.yaml
-rw-rw-rw- 1 codespace codespace 0 Jan 14 09:58 my-playbook.yaml
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

Add:

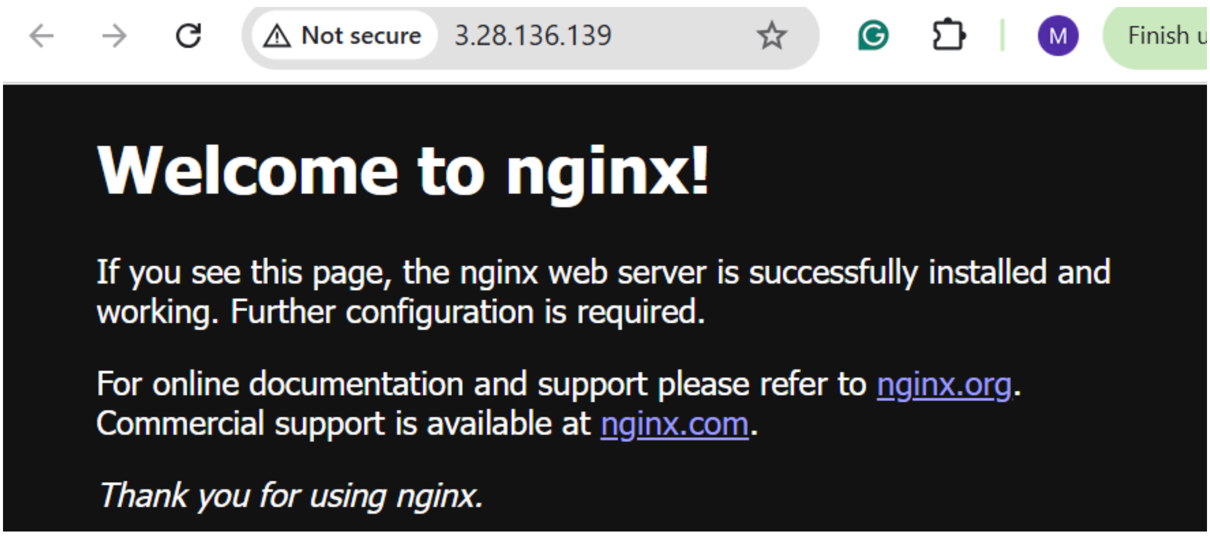
```
@manahilhabib031 →/workspaces/terraform_machine (main) $ nano my-playbook.yaml
@manahilhabib031 →/workspaces/terraform_machine (main) $ cat my-playbook.yaml
---
- name: Configure nginx web server
  hosts: ec2
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

5. **Run the playbook** on [ec2] group:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-play
book.yaml
[WARNING]: Host '3.28.134.239' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
ok: [3.28.134.239]
[WARNING]: Host '3.28.136.139' is using the discovered Python interpreter at '/usr/bin/pyth
on3.9', but future installation of another Python interpreter could cause a different inter
preter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendice
s/interpreter_discovery.html for more information.
ok: [3.28.136.139]

TASK [install nginx and update cache] ****************************************************
changed: [3.28.134.239]
changed: [3.28.136.139]

TASK [start nginx server] ****************************************************************
changed: [3.28.134.239]
changed: [3.28.136.139]

PLAY RECAP *******************************************************************************
3.28.134.239               : ok=3    changed=2    unreachable=0    failed=0    skipped=0
 rescued=0    ignored=0
3.28.136.139               : ok=3    changed=2    unreachable=0    failed=0    skipped=0
 rescued=0    ignored=0

@manahilhabib031 →/workspaces/terraform_machine (main) $
```

6. **Verify** nginx default page on [ec2] servers by visiting http://<public-ip-ec2>.



7. **Change target to droplet**:

Edit my-playbook.yaml to:

hosts: droplet

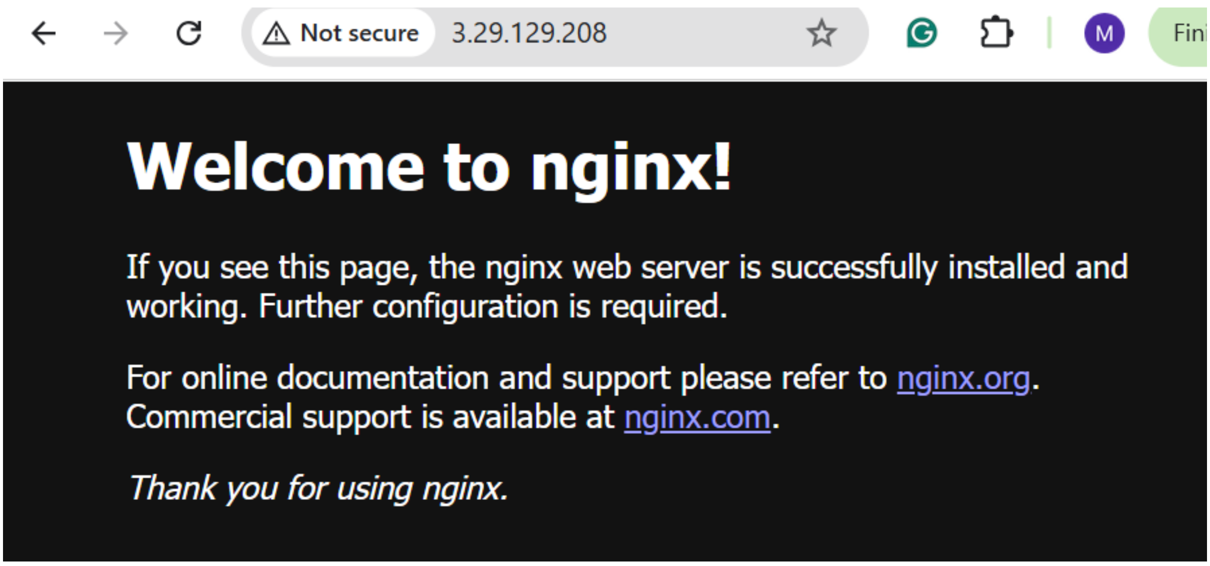
```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat my-playboo
k.yaml
---
- name: Configure nginx web server
  hosts: droplet
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: start nginx server
      service:
        name: nginx
        state: started
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

8. Re-run:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-pl
aybook.yaml

PLAY [Configure nginx web server] ***********************************************

TASK [Gathering Facts] **********************************************************
[WARNING]: Host '3.29.129.208' is using the discovered Python interpreter at '/usr/bin/py
thon3.9', but future installation of another Python interpreter could cause a different i
nterpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_app
endices/interpreter_discovery.html for more information.
ok: [3.29.129.208]

TASK [install nginx and update cache] *******************************************
changed: [3.29.129.208]

TASK [start nginx server] *******************************************************
changed: [3.29.129.208]

PLAY RECAP **********************************************************************
3.29.129.208               : ok=3    changed=2    unreachable=0    failed=0    skipped=0
   rescued=0    ignored=0

@manahilhabib031 →/workspaces/terraform_machine (main) $
```

9. Verify nginx default page on droplet:



Task 5 – Single nginx target group & HTTPS prerequisites

You will prepare **project-level Ansible configuration** and adjust nginx installation scope.

1. **Create project-level ansible.cfg** in repo root:

touch ansible.cfg

ls -la ansible.cfg

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ touch ansible.cfg
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ls -la ansible.cfg
-rw-rw-rw- 1 codespace codespace 0 Jan 14 10:09 ansible.cfg
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ 
```

Add:

[defaults]

host_key_checking=False

interpreter_python = /usr/bin/python3

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ nano ansible.cfg
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ cat ansible.cfg
[defaults]
host_key_checking=False
interpreter_python = /usr/bin/python3
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ 
```

2. **Switch Terraform EC2 count back to 1**:

In main.tf:

count = 1

```
  GNU nano 7.2                          main.tf
  vpc_id = aws_vpc.myapp_vpc.id
  subnet_cidr_block = var.subnet_cidr_block
  availability_zone = var.availability_zone
  env_prefix = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"
  env_prefix        = var.env_prefix
  instance_type     = var.instance_type
  availability_zone = var.availability_zone
  public_key        = var.public_key
  my_ip             = local.my_ip
  vpc_id            = aws_vpc.myapp_vpc.id
  subnet_id         = module.myapp-subnet.subnet.id

  # Loop count
  count           = 1
  # Use count.index to differentiate instances
  instance_suffix = count.index
}
```

3. Apply:

terraform apply -auto-approve

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ terraform apply -auto-approve
3dafbf03021d5b, 01m00s elapsed]
module.myapp-webserver[1].aws_instance.myapp-server: Still destroying... [id=i-00e0ab7ba73f09261, 01m00s elapsed]
module.myapp-webserver[1].aws_instance.myapp-server: Destruction complete after 1m0s
module.myapp-webserver[1].aws_security_group.web_sg: Destroying... [id=sg-0c6a5eb9e43f34532]
module.myapp-webserver[1].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-1]
module.myapp-webserver[2].aws_instance.myapp-server: Destruction complete after 1m0s
module.myapp-webserver[2].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-2]
module.myapp-webserver[2].aws_security_group.web_sg: Destroying... [id=sg-00a1e130ab16b5098]
module.myapp-webserver[1].aws_key_pair.ssh-key: Destruction complete after 0s
module.myapp-webserver[2].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-webserver[2].aws_security_group.web_sg: Destruction complete after 1s
module.myapp-webserver[1].aws_security_group.web_sg: Destruction complete after 1s

Apply complete! Resources: 0 added, 0 changed, 6 destroyed.

Outputs:

webserver_public_ips = [
  "3.28.134.239",
]
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

4. Outputs:

terraform output

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.134.239",
]
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

5. **Adjust hosts**:

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ nano hosts
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat hosts
[nginx]
3.28.134.239

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user

@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

6. **Update my-playbook.yaml**:

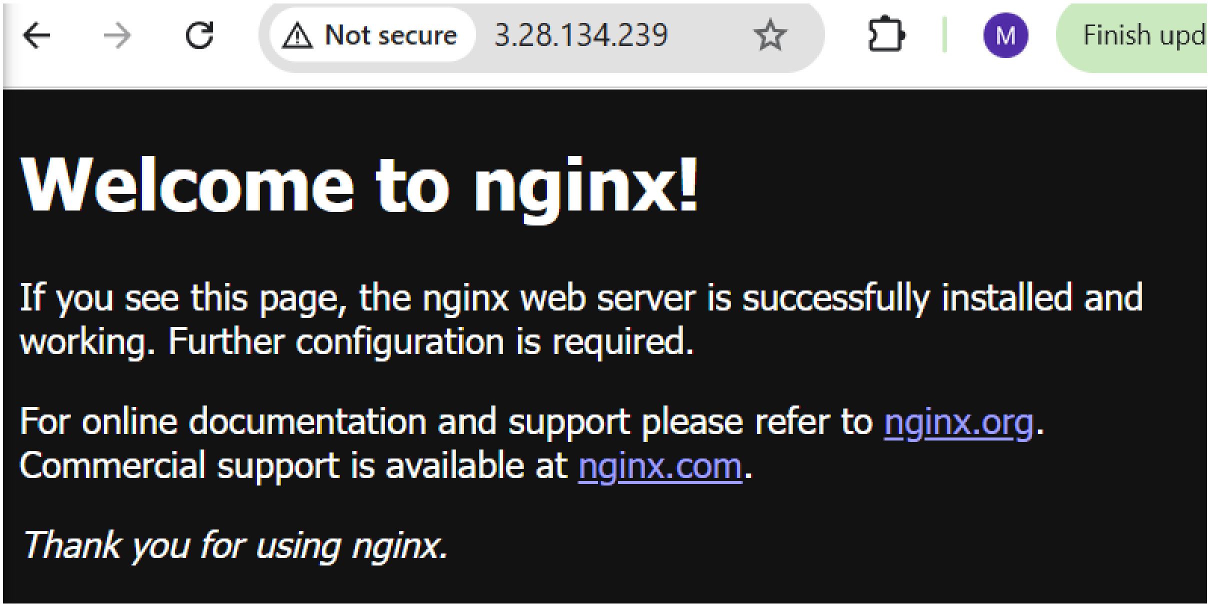
```
@manahilhabib031 →/workspaces/terraform_machine (main) $ nano my-playbook.yaml
@manahilhabib031 →/workspaces/terraform_machine (main) $ cat my-playbook.yaml
---
- name: Configure nginx web server
  hosts: nginx
  become: true
  tasks:
    - name: install nginx and update cache
      yum:
        name: nginx
        state: present
        update_cache: yes

    - name: install openssl
      yum:
        name: openssl
        state: present

    - name: start nginx server
      service:
        name: nginx
        state: started
        enabled: true

@manahilhabib031 →/workspaces/terraform_machine (main) $
```

7. **Run the playbook**:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ansible-playbook -i host
s my-playbook.yaml
//docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-wri
table-dir

PLAY [Configure nginx web server] ************************************************

TASK [Gathering Facts] ***********************************************************
[WARNING]: Host '3.28.134.239' is using the discovered Python interpreter at '/usr
/bin/python3.9', but future installation of another Python interpreter could cause
 a different interpreter to be discovered. See https://docs.ansible.com/ansible-co
re/2.20/reference_appendices/interpreter_discovery.html for more information.
ok: [3.28.134.239]

TASK [install nginx and update cache] ********************************************
ok: [3.28.134.239]

TASK [install openssl] ***********************************************************
ok: [3.28.134.239]

TASK [start nginx server] ********************************************************
changed: [3.28.134.239]

PLAY RECAP ***********************************************************************
3.28.134.239               : ok=4    changed=1    unreachable=0    failed=0    ski
pped=0    rescued=0    ignored=0

@manahilhabib031 →/workspaces/terraform_machine (main) $
```

8. **Verify nginx default page** at http://<public-ip>.



Task 6 - Ansible-managed SSL certificates

Extend your playbook to generate **self-signed SSL certificates** using dynamic public IP.

1. Append this play after the nginx play in my-playbook.yaml:

```
GNU nano 7.2                                my-playbook.yaml
      enabled: true

- name: Configure SSL certificates
  hosts: nginx
  become: true
  tasks:
    - name: Create SSL private directory
      file:
        path: /etc/ssl/private
        state: directory
        mode: '0700'

    - name: Create SSL certs directory
      file:
        path: /etc/ssl/certs
        state: directory
        mode: '0755'

    - name: Get IMDSv2 token
      uri:
        url: "http://169.254.169.254/latest/api/token"
        method: PUT
        headers:
          X-aws-ec2-metadata-token-ttl-seconds: "3600"
```

2. Run:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ansible-playbook -i host
s my-playbook.yaml

TASK [Create SSL private directory] *********************************************
changed: [3.28.134.239]

TASK [Create SSL certs directory] ***********************************************
changed: [3.28.134.239]

TASK [Get IMDSv2 token] *********************************************************
ok: [3.28.134.239]

TASK [Get current public IP] ****************************************************
ok: [3.28.134.239]

TASK [Show current public IP] ***************************************************
ok: [3.28.134.239] => {
    "msg": "Public IP: 3.28.134.239"
}

TASK [Generate self-signed SSL certificate] *************************************
changed: [3.28.134.239]

PLAY RECAP **********************************************************************
3.28.134.239               : ok=11   changed=3    unreachable=0    failed=0    ski
pped=0    rescued=0    ignored=0

@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

3. SSH and verify:

ssh ec2-user@<public-ip> -i ~/.ssh/id_ed25519

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ssh ec2-user@3.28.134.239 -i ~/.s
sh/id_ed25519
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Wed Jan 14 10:32:41 2026 from 4.240.39.192
[ec2-user@ip-10-0-10-62 ~]$
```

sudo cat /etc/ssl/certs/selfsigned.crt

```
[ec2-user@ip-10-0-10-62 ~]$ sudo cat /etc/ssl/certs/selfsigned.crt
-----BEGIN CERTIFICATE-----
MIIDPTCCAiWgAwIBAgIUJVvwNe7HreaXxUYh+Ny2Paz0NvcwDQYJKoZIhvcNAQEL
BQAwFzEVMBMGA1UEAwwMMy4yOC4xMzQuMjM5MB4XDTI2MDExNDEwMzI0MVoXDTI3
MDExNDEwMzI0MVowFzEVMBMGA1UEAwwMMy4yOC4xMzQuMjM5MIIBIjANBgkqhkiG
9w0BAQEFAAOCAQ8AMIIBCgKCAQEAt/Oef/mwj8tetrIst4nz69dy2ka/7Rgy1qGG
jcBMOxaYgvtZXtJsFtL87bgeOoACAkGXdQouMhh72hyqoBDeYjTiBtGjPkfZ0rV/
WG+HjI7sf9PkTV19LX9heWyPBHwKOibrMWMxlpmm7E9bSd8ish/Yx7AkkrjPWJI5
KEDJwYpWfKO/guh/NtTZWc0ImBPw/I2cyKslxd50dGBg6mwj3MRfuIsHKyTYd+H1
xktZLwnnNhYT8Q8iG0ezHQLlbRF34MBYxmB+Z82S5xeerKZZdFRVxJ6h7hPn2pEC
yBW+DIcwzrwulPNFcifQNW3jioze4JXo72K6LFHuR0YA9qoA8QIDAQABo4GAMH4w
HQYDVR0OBBYEFHUhEYw010Rit8Ph6fey9pb57icOMB8GA1UdIwQYMBaAFHUhEYw0
10Rit8Ph6fey9pb57icOMA8GA1UdEQQIMAaHBAMchu8wCQYDVR0TBAIwADALBgNV
HQ8EBAMCBaAwEwYDVR0lBAwwCgYIKwYBBQUHAwEwDQYJKoZIhvcNAQELBQADggEB
AAFkfR+JgFn33LktAcNxaMliMsDKqL8O/zx1smBYV42Rf4bj7sEaRaASbNoP3A6b
21USW874ASuXF7h62Ku/0uv5f5NSOGJt7Jh1OGlFgHhQXJXN2zfEzr7kJETyXPpG
tssXcGaSVMgPnGgePPN0yRJu9jpaj1J74i7nK6Lpk004DkK9Y3dm2SMABif1bnBu
kTEQVyXapNBwhozbjIvteZ9yPPOLCtCrwQ5mIJi+AaZ3WvZEGBhXU51T3ENXuZqJ
jWimVUK3DNuXxKv2Vf/T4NQ4M2l665rhcIJ4HFKJltitQFNnqHNjnH3TS0NKLZco
thdEONkp5KqoPGU4QOIT3Zo=
-----END CERTIFICATE-----
[ec2-user@ip-10-0-10-62 ~]$
```


sudo cat /etc/ssl/private/selfsigned.key


```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ssh ec2-user@3.28.134.239 -i ~/.s
sh/id_ed25519
HKqgEN5iNOIG0aM+R9nStX9Yb4eMjux/0+RNXX0tf2F5bI8EfAo6JusxYzGWmabs
T1tJ3yKyH9jHsCSSuM9YkjkoQMnBilZ8o7+C6H821NlZzQiYE/D8jZzIqyXF3nR0
YGDqbCPcxF+4iwcrJNh34fXGS1kvCec2FhPxDyIbR7MdAuVtEXfgwFjGYH5nzZLn
F56spll0VFXEnqHuE+fakQLIFb4MhzDOvC6U80VyJ9A1beOKjN7glejvYrosUe5H
RgD2qgDxAgMBAAECggEADHLjuJnEvaPP4pb3Q2qGGTao5/sJamRiAw1xBi8hLuSD
aLsq2/8V3+DidhH+QGDrrIxqJrphevQWOaIJWAYcOVLb3bCalrs/vgHuFCWOwPqR
KGtwvrLuJuyWbk8O9LOSRkeI+3wmMyDDNP3VLzrMZc1ARe9L41QN0DArxxIaCqC9
LnS/EEViZWt0COBscojwSW5Qf5H6WIi7Ez0LB9mIhnP2p51SUvQzH0+LH+ByZwUX
SfR66uycvbe4HXvzPGE7FogR5c7qnGPwa9KEEl1UtddX8+JUDv8ng2i/WqrGqkEn
p9EaDUm5Lx7fk4cAOzwbN0dOr+bUSADa4I8iODbSGwKBgQDch/dDg40T4jkk5CTn
Ex9Hh1Gu51wznjrqOFJB7JVuMixR/6EjS3cfrvz3TfJnD0xLTgRRtmxudLAzrjLp
FR2qi/r3qftiuhxX2dDXXliz9uNTb9/Dok75QV3qnf93Fii++IkBIFwH7kYA5Rta
+uCsDAPn+LCAcmEDu5b+AbwYOwKBgQDViYx3tLfIGUBa8ecL+33HE3qFhLxXOQSU
wEk2GHahqEugHfvMLFb0MqD+T6j1ioE0uAfHyQytOYf9gH2J3WKDB5vc94W0o6pQ
qaeQC59dXjE68RMumtpxI6fvMBKkEFMeDptZ9GWcpcUyrPqDlWpj8pB8W6X84R2l
mAaMPn3kwwKBgELnFMJspF4F4WQpunBHwtOGX+d51OTr+d+YmovZZDP8EKoHrTDx
8f+wrJmyEfnmEDVAQ9suRpLz34iuLk+70jYtJUQBU3xODc8xC0RrIOnn2mszo2TM
Vr0TcZR9vkS4ekAFUjr/QCw/wwj+eGAPgB1xv7WeIp21FW12eYYxrIiFAoGBALbc
VXflQESQqzW2Vb9DHFCwc4mHQ6oF39BqYcd7Z4vEDQ8O3iPYDJaODKP7H5s1grPc
rHz/xGVLERObRuYauVOv3l95b/xi82SkBCQsb8qK8OmirGSKo2mjuRnxNuVcPXFN
3OdJjQJ6Ssa8rsKj3zm77NMxwJEvTi4SY/kRQsn1AoGAGeTZNu2n6ISzm93NBl6b
6wyizPPKLTgW3UHitp8n2D75KPF+n9JCV2C9Yco3hdhFRILpbq4+pmwiRHa6OgMy
uF6XY8uL6ZWAFSmJsZuu3wYapsESl8EBpIj7EK1n1H0yEesuXXUpf9qfdu4w+Ylw
yU6Wz6GIY2XjDFbyNsf+EMs=
-----END PRIVATE KEY-----
[ec2-user@ip-10-0-10-62 ~]$
```


Exit

```
[ec2-user@ip-10-0-10-62 ~]$ exit
logout
Connection to 3.28.134.239 closed.
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

Task 7 - PHP front-end deployment with templates

Deploy a PHP application and Nginx configuration using Ansible copy and template.

1. **Create directories and files**:

mkdir -p files templates

touch files/index.php

touch templates/nginx.conf.j2

ls -R

```
./aws/dist/setuptools/_vendor:
jaraco

./aws/dist/setuptools/_vendor/jaraco:
text

./aws/dist/setuptools/_vendor/jaraco/text:
'Lorem ipsum.txt'

./aws/dist/wheel-0.45.1.dist-info:
INSTALLER    METADATA  REQUESTED  direct_url.json
LICENSE.txt  RECORD    WHEEL      entry_points.txt

./files:
index.php

./modules:
subnet  webserver

./modules/subnet:
main.tf  outputs.tf  variables.tf

./modules/webserver:
main.tf  outputs.tf  variables.tf

./templates:
nginx.conf.j2
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

2. **Fill files/index.php** with the following PHP metadata page:

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat files/index.php
</head>
<body>
    <div class="container">
        <h1>🚀 Nginx Front End Web Server </h1>

        <div class="info"><span class="label">Hostname:</span> <?= htmlspecialchars($ho
stname) ?></div>
        <div class="info"><span class="label">Instance ID:</span> <?= htmlspecialchars(
$instance_id) ?></div>
        <div class="info"><span class="label">Private IP:</span> <?= htmlspecialchars($
private_ip) ?></div>
        <div class="info"><span class="label">Public IP:</span> <?= htmlspecialchars($p
ublic_ip) ?></div>
        <div class="info"><span class="label">Public DNS:</span>
            <a href="https://<?= htmlspecialchars($public_dns) ?>" target="_blank">
            https://<?= htmlspecialchars($public_dns) ?></a>
        </div>
        <div class="info"><span class="label">Deployed:</span> <?= $deployed_date ?></d
iv>
        <div class="info"><span class="label">Status:</span> ✅ Active and Running</div
>
        <div class="info"><span class="label">Managed By:</span> Terraform + Ansible</d
iv>
    </div>
</body>
</html>
```

3. **Fill templates/nginx.conf.j2** with the following Nginx configuration template:

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat templates/nginx.conf.j2
        server_name {{ server_public_ip }};
        ssl_certificate /etc/ssl/certs/selfsigned.crt;
        ssl_certificate_key /etc/ssl/private/selfsigned.key;

        location / {
            root /usr/share/nginx/html;
            index index.php index.html index.htm;
    #       proxy_pass http://158.252.94.241:80;
    #       proxy_pass http://backend_servers;

            # 🔴 This block is necessary for Php Website
            location ~ \.php$ {
                include fastcgi_params;
                fastcgi_pass unix:/run/php-fpm/www.sock;
                fastcgi_index index.php;
                fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
            }
        }
    }

    server {
        listen 80;
        server_name _;
        return 301 https://$host$request_uri;
    }
}
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

4. **Add this play to my-playbook.yaml**:

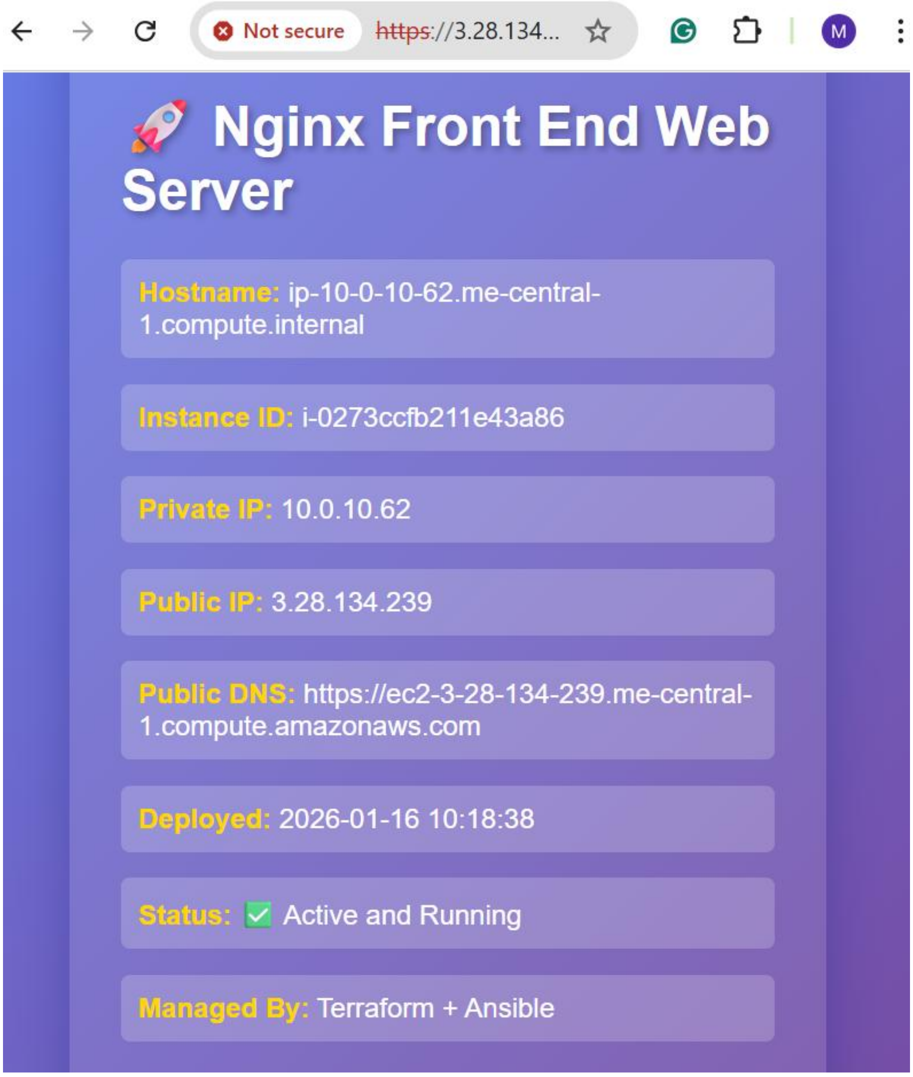
```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ cat my-playbook.yaml
    - name: Copy website files
      copy:
        src: files/index.php
        dest: /usr/share/nginx/html/index.php
        owner: nginx
        group: nginx
        mode: '0644'

    - name: Copy nginx.conf template
      template:
        src: templates/nginx.conf.j2
        dest: /etc/nginx/nginx.conf
        owner: root
        group: root
        mode: '0644'

    - name: Restart nginx
      service:
        name: nginx
        state: restarted

    - name: Start and enable php-fpm
      service:
        name: php-fpm
        state: started
        enabled: true
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

5. **Run the playbook**:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
TASK [Get current public IP] ************************************************************
ok: [3.28.134.239]

TASK [Set server public IP fact] ********************************************************
ok: [3.28.134.239]

TASK [install php-fpm and php-curl] *****************************************************
ok: [3.28.134.239]

TASK [Copy website files] ***************************************************************
ok: [3.28.134.239]

TASK [Copy nginx.conf template] *********************************************************
changed: [3.28.134.239]

TASK [Restart nginx] ********************************************************************
changed: [3.28.134.239]

TASK [Start and enable php-fpm] *********************************************************
changed: [3.28.134.239]

PLAY RECAP ******************************************************************************
3.28.134.239               : ok=9    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

6. Visit https://<public-ip>:
    - Accept SSL warning.
    - Verify the PHP page content.



Task 8 – Docker & Docker Compose provisioning via Ansible

Deploy Docker and Docker Compose on a new EC2 instance.

1. **Destroy old infrastructure**:

terraform destroy -auto-approve

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ terraform destroy -auto-appro
ve
module.myapp-webserver[0].aws_instance.myapp-server: Destroying... [id=i-0273ccfb211e43
a86]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-043befbae7aaf
ba58]
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0273ccfb
211e43a86, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-043befb
ae7aafba58, 00m10s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 17s
module.myapp-webserver[0].aws_instance.myapp-server: Still destroying... [id=i-0273ccfb
211e43a86, 00m20s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Destruction complete after 30s
module.myapp-webserver[0].aws_key_pair.ssh-key: Destroying... [id=dev-serverkey-0]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0c935a5ae1e887a
db]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-06e22018b4e66
43ba]
module.myapp-webserver[0].aws_key_pair.ssh-key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0c62b9ca498e486e4]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

2. **Recreate fresh infrastructure** (1 instance):

terraform apply -auto-approve

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform apply -auto-approve

module.myapp-subnet.aws_internet_gateway.myapp_igw: Creating...
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creating...
module.myapp-webserver[0].aws_security_group.web_sg: Creating...
module.myapp-subnet.aws_internet_gateway.myapp_igw: Creation complete after 0s [id=igw-062cf3ff00434be97]
module.myapp-subnet.aws_default_route_table.main_rt: Creating...
module.myapp-subnet.aws_default_route_table.main_rt: Creation complete after 1s [id=rtb-0d9507f8f006606ac]
module.myapp-webserver[0].aws_security_group.web_sg: Creation complete after 3s [id=sg-04e88e364761c0bd3]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Still creating... [00m10s elapsed]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Creation complete after 11s [id=subnet-0edcceac81849a173]
module.myapp-webserver[0].aws_instance.myapp-server: Creating...
module.myapp-webserver[0].aws_instance.myapp-server: Still creating... [00m10s elapsed]
module.myapp-webserver[0].aws_instance.myapp-server: Creation complete after 13s [id=i-020df23aaf4d95c7d]

Apply complete! Resources: 7 added, 0 changed, 0 destroyed.

Outputs:

webserver_public_ips = [
  "3.28.187.42",
]
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

terraform output

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform output
webserver_public_ips = [
  "3.28.187.42",
]
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

3. **Update hosts**:

```
  GNU nano 7.2                          hosts
[nginx]
3.28.134.239

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user

[docker_servers]
3.28.134.239

[docker_servers:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user
```

4. **Replace my-playbook.yaml** content with:

```
GNU nano 7.2                          my-playbook.yaml *
---
- name: Configure Docker
  hosts: all
  become: true
  tasks:
    - name: install docker and update cache
      yum:
        name: docker
        state: present
        update_cache: yes

    - name: start and enable docker
      service:
        name: docker
        state: started
        enabled: true

- name: Install Docker Compose
  hosts: all
  become: true
  gather_facts: true
  tasks:
    - name: create docker cli-plugins directory
```

5. **Run the play**:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
ok: [3.28.187.42]

TASK [create docker cli-plugins directory] *******************************************
changed: [3.28.187.42]

TASK [install docker-compose] ********************************************************
changed: [3.28.187.42]

TASK [View architecture of the system] ***********************************************
ok: [3.28.187.42] => {
    "msg": "System architecture of 3.28.187.42 is x86_64"
}

TASK [Alternate method to view architecture of the system] ***************************
ok: [3.28.187.42] => {
    "msg": "System architecture of 3.28.187.42 is x86_64"
}

TASK [restart docker service] ********************************************************
changed: [3.28.187.42]

PLAY RECAP ***************************************************************************
3.28.187.42                : ok=9    changed=5    unreachable=0    failed=0    skipped=
0    rescued=0    ignored=0

@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

6. **Verify Docker**:

ssh ec2-user@<public-ip> -i ~/.ssh/id_ed25519

sudo docker ps

exit

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ssh ec2-user@3.28.187.42 -i ~
/.ssh/id_ed25519
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Jan 16 10:38:06 2026 from 20.192.21.51
[ec2-user@ip-10-0-10-244 ~]$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
[ec2-user@ip-10-0-10-244 ~]$ oexit
-bash: oexit: command not found
[ec2-user@ip-10-0-10-244 ~]$ exit
logout
Connection to 3.28.187.42 closed.
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Task 9 – Gitea Docker stack via Ansible + Terraform security group update

Run containers for **Gitea + Postgres** and open port 3000 in the security group.

1. **Extend my-playbook.yaml** by appending:

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ cat my-playbook.yaml
  become: true
  vars_files:
    - project-vars.yaml
  tasks:
    - name: add user to docker group
      user:
        name: "{{ normal_user }}"
        groups: docker
        append: yes

    - name: reconnect to apply group changes
      meta: reset_connection

    - name: verify docker access
      command: docker ps
      register: docker_ps
      changed_when: false

    - name: display docker ps output
      debug:
        var: docker_ps.stdout

    - name: fail if docker is not accessible
      fail:
        msg: "Docker is not accessible on this host"
      when: docker_ps.rc != 0
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

2. **Create project-vars.yaml**:

touch project-vars.yaml

Add:

normal_user: ec2-user

docker_compose_file_location: <location-of-file>

```
 GNU nano 7.2                      project-vars.yaml *
normal_user: ec2-user
docker_compose_file_location: /workspaces/terraform_machine
```

3. **Add the deploy containers play**:

```
 GNU nano 7.2                      my-playbook.yaml *
- name: Deploy Docker Containers
  hosts: all
  become: true
  user: "{{ normal_user }}"
  vars_files:
    - project-vars.yaml
  tasks:
    - name: check if docker-compose file exists
      stat:
        path: /home/{{ normal_user }}/compose.yaml
      register: compose_file

    - name: copy docker-compose file
      copy:
        src: "{{ docker_compose_file_location }}/compose.yaml"
        dest: /home/{{ normal_user }}/compose.yaml
        mode: '0644'
      when: not compose_file.stat.exists

    - name: deploy containers using docker-compose
      command: docker compose up -d
      register: compose_result
      changed_when: "'Creating' in compose_result.stdout or 'Recreating' in compose_re>
```

4. **Create compose.yaml** in repo root:

```
GNU nano 7.2                              compose.yaml *
services:
  gitea:
    image: gitea/gitea:latest
    container_name: gitea
    environment:
      - DB_TYPE=postgres
      - DB_HOST=db:5432
      - DB_NAME=gitea
      - DB_USER=gitea
      - DB_PASSWD=gitea
    restart: always
    volumes:
      - gitea:/data
    ports:
      - 3000:3000
    extra_hosts:
      - "www.jenkins.com:host-gateway"
    networks:
      - webnet
  db:
    image: postgres:alpine
    container_name: gitea_db
    environment:
      - POSTGRES_USER=gitea
```

5. **Run playbook** and visit the public-ip on your browser but webpage will not be shown due to permission issue:

ansible-playbook -i hosts my-playbook.yaml

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-playbook -i hosts my-playbook.yaml
    "docker_ps.stdout": "CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS
   NAMES"
}

TASK [fail if docker is not accessible] ********************************************
skipping: [3.28.187.42]

PLAY [Deploy Docker Containers] ****************************************************

TASK [Gathering Facts] *************************************************************
ok: [3.28.187.42]

TASK [check if docker-compose file exists] *****************************************
ok: [3.28.187.42]

TASK [copy docker-compose file] ****************************************************
changed: [3.28.187.42]

TASK [deploy containers using docker-compose] **************************************
ok: [3.28.187.42]

PLAY RECAP *************************************************************************
3.28.187.42                : ok=17   changed=3    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

6. **Update security group** in modules/webserver/main.tf to add port 3000:

ingress {

from_port = 3000

to_port = 3000

protocol = "tcp"

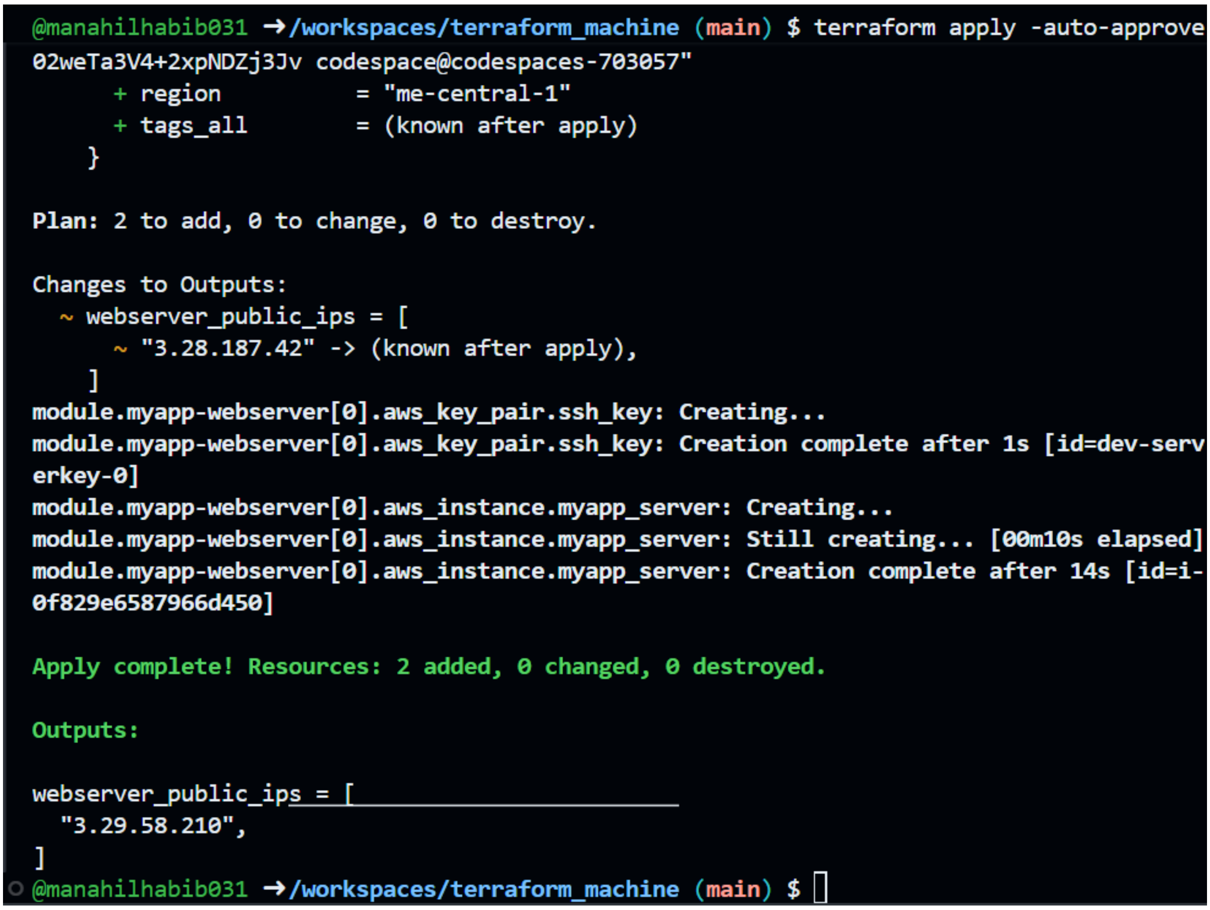
cidr_blocks = ["0.0.0.0/0"]

}

```
ingress {
  from_port   = 3000
  to_port     = 3000
  protocol    = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
}
```

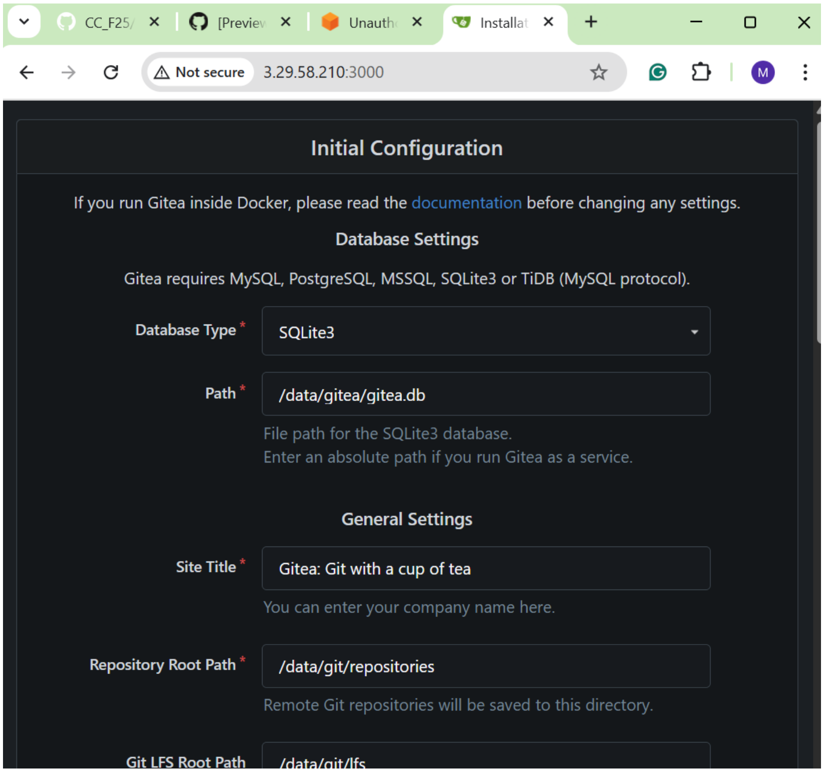
7. Apply:

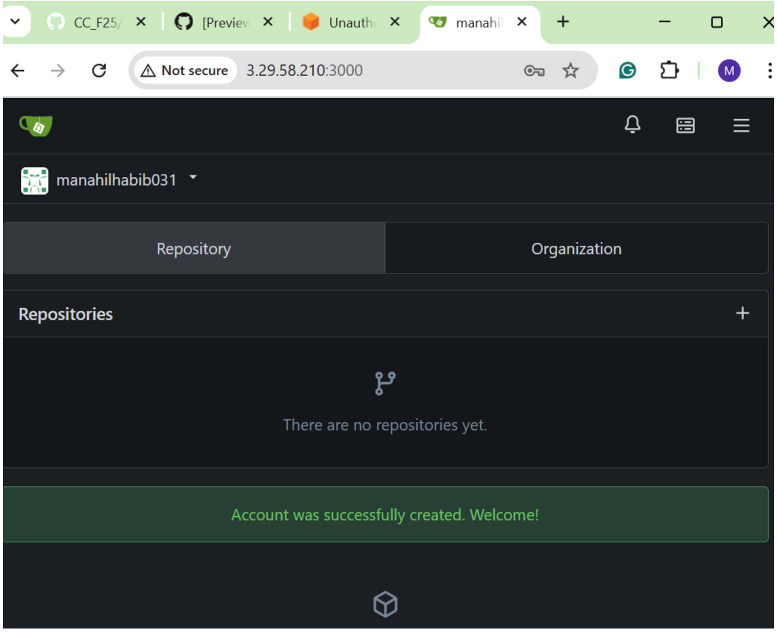
terraform apply -auto-approve



8. **Access Gitea**:

Visit http://<public-ip>:3000.





Task 10 – Automating Ansible with Terraform (null_resource)

Have Terraform trigger Ansible automatically after EC2 creation.

1. **Add null_resource** to main.tf:

```
  GNU nano 7.2                          main.tf *
  # Use count.index to differentiate instances
  instance_suffix = count.index
}

resource "null_resource" "configure_server" {
  triggers = {
    # Join all public IPs of your webserver module into a comma-separated string
    webserver_public_ips_for_ansible = join(",", [for i in module.myapp-webserver : >
  }

  depends_on = [
    module.myapp-webserver
  ]

  provisioner "local-exec" {
    command = <<EOT
ansible-playbook -i ${self.triggers.webserver_public_ips_for_ansible}, \
--private-key "${var.private_key}" \
--user ec2-user \
my-playbook.yaml
EOT
  }
}
```

2. **Destroy and recreate** infrastructure:

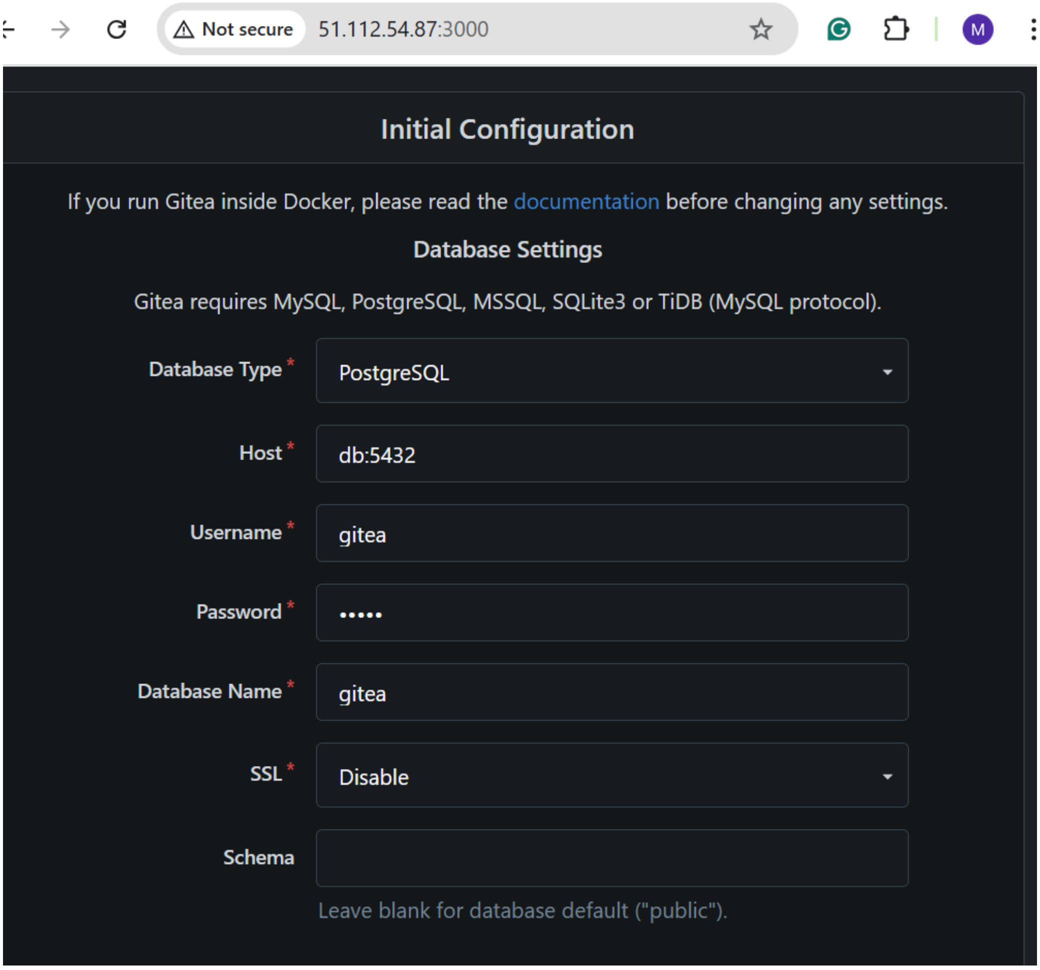
terraform destroy -auto-approve

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform destroy -auto-ap
rove
3ff00434be97, 00m30s elapsed]
module.myapp-webserver[0].aws_instance.myapp_server: Still destroying... [id=i-0f829
6587966d450, 00m40s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-062c
3ff00434be97, 00m40s elapsed]
module.myapp-webserver[0].aws_instance.myapp_server: Still destroying... [id=i-0f829
6587966d450, 00m50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Still destroying... [id=igw-062c
3ff00434be97, 00m50s elapsed]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 1m0s
module.myapp-webserver[0].aws_instance.myapp_server: Still destroying... [id=i-0f829
6587966d450, 01m00s elapsed]
module.myapp-webserver[0].aws_instance.myapp_server: Destruction complete after 1m3s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0edcceac8184
a173]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-096419f829
ae91db]
module.myapp-webserver[0].aws_key_pair.ssh_key: Destroying... [id=dev-serverkey-0]
module.myapp-webserver[0].aws_key_pair.ssh_key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-092ebdb60b819b216]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 7 destroyed.
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

terraform apply -auto-approve

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform apply -auto-appro
ve
null_resource.configure_server (local-exec): ok: [51.112.54.87]

null_resource.configure_server (local-exec): PLAY [Wait for SSH to be ready] ********
*****************************************

null_resource.configure_server (local-exec): TASK [Wait 300 seconds for port 22 to be
 open] *********************************
null_resource.configure_server: Still creating... [02m10s elapsed]
null_resource.configure_server (local-exec): ok: [51.112.54.87 -> localhost]

null_resource.configure_server (local-exec): PLAY RECAP *****************************
*****************************************
null_resource.configure_server (local-exec): 51.112.54.87              : ok=18   cha
nged=7    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

null_resource.configure_server: Creation complete after 2m11s [id=4532837227895864621
]

Apply complete! Resources: 1 added, 0 changed, 1 destroyed.

Outputs:

webserver_public_ips = [
  "51.112.54.87",
]
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

5. Verify Gitea / Nginx application is reachable at the appropriate URL.



Task 11 – Dynamic inventory with aws_ec2 plugin

Let Ansible discover EC2 instances dynamically via the aws_ec2 inventory plugin.

1. **Update ansible.cfg**:

```
 GNU nano 7.2                     ansible.cfg *
[defaults]
host_key_checking = False
interpreter_python = /usr/bin/python3
deprecation_warnings = False
enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519
```

2. **Create inventory_aws_ec2.yaml**:

touch inventory_aws_ec2.yaml

ls -la inventory_aws_ec2.yaml

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ touch inventory_aws_ec2.yaml
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ ls -la inventory_aws_ec2.yam
l
-rw-rw-rw- 1 codespace codespace 0 Jan 18 08:08 inventory_aws_ec2.yaml
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Add initial content:

---

plugin: aws_ec2

regions:

- me-central-1

```
  GNU nano 7.2                    inventory_aws_ec2.yaml *
---
plugin: aws_ec2
regions:
  - me-central-1
```

3. **Ensure Terraform code** includes both dev and prod webservers in **main.tf**:

```
  GNU nano 7.2                              main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block           = var.vpc_cidr_block
  enable_dns_hostnames = true
  tags = {
     Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source                 = "./modules/subnet"
  vpc_id                 = aws_vpc.myapp_vpc.id
  subnet_cidr_block      = var.subnet_cidr_block
  availability_zone      = var.availability_zone
  env_prefix             = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source            = "./modules/webserver"

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Locati
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To
```

4. **Add outputs in outputs.tf**:

```
  GNU nano 7.2                              outputs.tf *
output "webserver_public_ips" {
  value = [for i in module.myapp-webserver : i.public_ip]
}

output "prod-webserver_public_ips" {
  value = [for i in module.myapp-webserver-prod : i.public_ip]
}
```

5. **Rebuild infra**:

terraform init

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ terraform init
Initializing the backend...
Initializing modules...
- myapp-webserver-prod in modules/webserver
Initializing provider plugins...
- Reusing previous version of hashicorp/http from the dependency lock file
- Reusing previous version of hashicorp/aws from the dependency lock file
- Reusing previous version of hashicorp/null from the dependency lock file
- Using previously-installed hashicorp/http v3.5.0
- Using previously-installed hashicorp/aws v6.28.0
- Using previously-installed hashicorp/null v3.2.4

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

terraform apply -auto-approve

```
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $ terraform apply -auto-appro
e
    ]
null_resource.configure_server: Destroying... [id=4532837227895864621]
null_resource.configure_server: Destruction complete after 0s
module.myapp-webserver-prod[0].aws_key_pair.ssh_key: Creating...
module.myapp-webserver-prod[0].aws_security_group.web_sg: Creating...
module.myapp-webserver-prod[0].aws_key_pair.ssh_key: Creation complete after 1s [id=p
od-serverkey-0]
module.myapp-webserver-prod[0].aws_security_group.web_sg: Creation complete after 4s
id=sg-0f92a1341c890c281]
module.myapp-webserver-prod[0].aws_instance.myapp_server: Creating...
module.myapp-webserver-prod[0].aws_instance.myapp_server: Still creating... [00m10s e
apsed]
module.myapp-webserver-prod[0].aws_instance.myapp_server: Creation complete after 14s
[id=i-05eaa8131fced154d]

Apply complete! Resources: 3 added, 0 changed, 1 destroyed.

Outputs:

prod-webserver_public_ips = [
  "158.252.83.205",
]
webserver_public_ips = [
  "51.112.54.87",
]
@manahilhabib031 ➜ /workspaces/terraform_machine (main) $
```

terraform output

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ terraform output
prod-webserver_public_ips = [
  "158.252.83.205",
]
webserver_public_ips = [
  "51.112.54.87",
]
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

6. **Install boto3 and botocore**:

$(which python) -m pip install boto3 botocore

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ $(which python) -m pip insta
ll boto3 botocore
Collecting boto3
  Downloading boto3-1.42.30-py3-none-any.whl.metadata (6.8 kB)
Collecting botocore
  Downloading botocore-1.42.30-py3-none-any.whl.metadata (5.9 kB)
Collecting jmespath<2.0.0,>=0.7.1 (from boto3)
  Downloading jmespath-1.0.1-py3-none-any.whl.metadata (7.6 kB)
Collecting s3transfer<0.17.0,>=0.16.0 (from boto3)
  Downloading s3transfer-0.16.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /home/codespace/.local/l
ib/python3.12/site-packages (from botocore) (2.9.0.post0)
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /home/codespace/.local/li
b/python3.12/site-packages (from botocore) (2.5.0)
Requirement already satisfied: six>=1.5 in /home/codespace/.local/lib/python3.12/site-
packages (from python-dateutil<3.0.0,>=2.1->botocore) (1.17.0)
Downloading boto3-1.42.30-py3-none-any.whl (140 kB)
Downloading botocore-1.42.30-py3-none-any.whl (14.6 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 14.6/14.6 MB 29.2 MB/s  0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.16.0-py3-none-any.whl (86 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.42.30 botocore-1.42.30 jmespath-1.0.1 s3transfer-0.16.0
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

- Verify the version

$(which python) -c "import boto3, botocore; print(boto3.__version__)"

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ $(which python) -c "import b
oto3, botocore; print(boto3.__version__)"
1.42.30
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

7. **Check inventory graph**:

ansible-inventory -i inventory_aws_ec2.yaml –graph

```
@manahilhabib031 →/workspaces/terraform_machine (main) $ ansible-inventory -i invento
ry_aws_ec2.yaml --graph

Failed to parse inventory with 'yaml' plugin.

<<< caused by >>>

Plugin configuration YAML file, not YAML inventory
Origin: <inventory plugin 'yaml' with source '/workspaces/terraform_machine/inventory_
aws_ec2.yaml'>

[WARNING]: Failed to parse inventory with 'ini' plugin: Failed to parse inventory: Inv
alid host pattern '---' supplied, '---' is normally a sign this is a YAML file.

Failed to parse inventory with 'ini' plugin.

<<< caused by >>>

Failed to parse inventory: Invalid host pattern '---' supplied, '---' is normally a si
gn this is a YAML file.
Origin: /workspaces/terraform_machine/inventory_aws_ec2.yaml

[WARNING]: Unable to parse /workspaces/terraform_machine/inventory_aws_ec2.yaml as an
inventory source
[WARNING]: No inventory was parsed, only implicit localhost is available
@all:
  |--@ungrouped:
@manahilhabib031 →/workspaces/terraform_machine (main) $
```

Task 12 – Filtering EC2 instances by tags & instance type

Augment the inventory plugin to group by tags and instance type, then limit plays to specific groups.

1. **Modify inventory_aws_ec2.yaml** to add tag-based grouping:

```
 GNU nano 7.2                    inventory_aws_ec2.yaml
---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"
```

Check graph:

ansible-inventory -i inventory_aws_ec2.yaml –graph

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-inventory -i inventory_aws_ec2.yaml --graph
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: Deprecation warnings can be disabled by setting `deprecation_warnings=False` in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is deprecated. This feature will be removed from ansible-core version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' is deprecated. This feature will be removed from ansible-core version 2.24. Use ansible.module_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This feature will be removed from ansible-core version 2.23.
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |  |--ec2-158-252-7-15.me-central-1.compute.amazonaws.com
  |  |--ec2-158-252-83-205.me-central-1.compute.amazonaws.com
  |  |--ec2-51-112-54-87.me-central-1.compute.amazonaws.com
  |--@tag_Name_web:
  |  |--ec2-158-252-7-15.me-central-1.compute.amazonaws.com
  |--@tag_Name_prod_ec2_instance_0:
  |  |--ec2-158-252-83-205.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_ec2_instance_0:
  |  |--ec2-51-112-54-87.me-central-1.compute.amazonaws.com
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

2. **Extend to group by instance type as well**:

```
GNU nano 7.2                    inventory_aws_ec2.yaml *
---
plugin: aws_ec2
regions:
  - me-central-1

keyed_groups:
  - key: tags
    prefix: tag
    separator: "_"

  - key: instance_type
    prefix: instance_type
    separator: "_"
```

Check graph again:

ansible-inventory -i inventory_aws_ec2.yaml –graph

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-inven
tory -i inventory_aws_ec2.yaml --graph
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is deprec
ated. This feature will be removed from ansible-core version 2.24. Use ansible.module_
utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' is depr
ecated. This feature will be removed from ansible-core version 2.24. Use ansible.modul
e_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This fea
ture will be removed from ansible-core version 2.23.
@all:
  |--@ungrouped:
  |--@aws_ec2:
  |  |--ec2-158-252-7-15.me-central-1.compute.amazonaws.com
  |  |--ec2-158-252-83-205.me-central-1.compute.amazonaws.com
  |  |--ec2-51-112-54-87.me-central-1.compute.amazonaws.com
  |--@tag_Name_web:
  |  |--ec2-158-252-7-15.me-central-1.compute.amazonaws.com
  |--@instance_type_t3_micro:
  |  |--ec2-158-252-7-15.me-central-1.compute.amazonaws.com
  |  |--ec2-51-112-54-87.me-central-1.compute.amazonaws.com
  |--@tag_Name_prod_ec2_instance_0:
  |  |--ec2-158-252-83-205.me-central-1.compute.amazonaws.com
  |--@instance_type_t3_nano:
  |  |--ec2-158-252-83-205.me-central-1.compute.amazonaws.com
  |--@tag_Name_dev_ec2_instance_0:
  |  |--ec2-51-112-54-87.me-central-1.compute.amazonaws.com
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

3. **Prepare my-playbook.yaml** for nginx+SSL+PHP on hosts: all:

```
GNU nano 7.2                    my-playbook.yaml *
---
- name: Configure nginx web server
  hosts: all
  become: true
  tasks:
  - name: install nginx and update cache
    yum:
      name: nginx
      state: present
      update_cache: yes

  - name: install openssl
    yum:
      name: openssl
      state: present

  - name: start nginx server
    service:
      name: nginx
      state: started
      enabled: true

- name: Configure SSL certificates
  hosts: all

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

4. **Run on all instances**:

ansible-playbook -i inventory_aws_ec2.yaml my-playbook.yaml

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible -i in
ventory_aws_ec2.yaml all -m ping
` in ansible.cfg.
[DEPRECATION WARNING]: Importing 'to_text' from 'ansible.module_utils._text' is deprec
ated. This feature will be removed from ansible-core version 2.24. Use ansible.module_
utils.common.text.converters instead.
[DEPRECATION WARNING]: Importing 'to_native' from 'ansible.module_utils._text' is depr
ecated. This feature will be removed from ansible-core version 2.24. Use ansible.modul
e_utils.common.text.converters instead.
[DEPRECATION WARNING]: Passing `disable_lookups` to `template` is deprecated. This fea
ture will be removed from ansible-core version 2.23.
[WARNING]: Found variable using reserved name 'tags'.
Origin: <unknown>

tags

[WARNING]: Host 'public_dns_name' is using the discovered Python interpreter at '/usr/
bin/python3.9', but future installation of another Python interpreter could cause a di
fferent interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/r
eference_appendices/interpreter_discovery.html for more information.
public_dns_name | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

5. **Run only on dev instances**:

ansible-playbook -i inventory_aws_ec2.yaml -l tag_Name_dev_* my-playbook.yaml

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-playb
ook -i inventory_aws_ec2.yaml -l tag_Name_dev_* my-playbook.yaml

PLAY [Deploy Nginx website and configuration files] **********************************

TASK [Gathering Facts] ***************************************************************
ok: [public_dns_name]

TASK [install php-fpm and php-curl] **************************************************
changed: [public_dns_name]

TASK [Copy website files] ************************************************************
changed: [public_dns_name]

TASK [Copy nginx.conf template] ******************************************************
changed: [public_dns_name]

TASK [Restart nginx] *****************************************************************
changed: [public_dns_name]

TASK [Start and enable php-fpm] ******************************************************
changed: [public_dns_name]

PLAY RECAP ***************************************************************************
public_dns_name            : ok=18   changed=10   unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0

(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

6. **Run only on prod instances**:

ansible-playbook -i inventory_aws_ec2.yaml -l tag_Name_prod_* my-playbook.yaml

```
(ansible-core) @manahilhabib031 →/workspaces/terraform_machine (main) $ ansible-playb
ook -i inventory_aws_ec2.yaml -l tag_Name_prod_* my-playbook.yaml

PLAY [Deploy Nginx website and configuration files] ***********************************

TASK [Gathering Facts] ****************************************************************
ok: [public_dns_name]

TASK [install php-fpm and php-curl] ***************************************************
ok: [public_dns_name]

TASK [Copy website files] *************************************************************
ok: [public_dns_name]

TASK [Copy nginx.conf template] *******************************************************
ok: [public_dns_name]

TASK [Restart nginx] ******************************************************************
changed: [public_dns_name]

TASK [Start and enable php-fpm] *******************************************************
ok: [public_dns_name]

PLAY RECAP ****************************************************************************
public_dns_name            : ok=18   changed=1    unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0

(ansible-core) @manahilhabib031 →/workspaces/terraform_machine (main) $
```

7. **Run only on t3.micro** instances:

ansible-playbook -i inventory_aws_ec2.yaml -l instance_type_t3_micro my-playbook.yaml

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml -l instance_type_t3_micro my-playbook.yaml

PLAY [Deploy Nginx website and configuration files] ***********************************

TASK [Gathering Facts] ****************************************************************
ok: [public_dns_name]

TASK [install php-fpm and php-curl] ***************************************************
ok: [public_dns_name]

TASK [Copy website files] *************************************************************
ok: [public_dns_name]

TASK [Copy nginx.conf template] *******************************************************
ok: [public_dns_name]

TASK [Restart nginx] ******************************************************************
changed: [public_dns_name]

TASK [Start and enable php-fpm] *******************************************************
ok: [public_dns_name]

PLAY RECAP ****************************************************************************
public_dns_name            : ok=18   changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

8. **Run only on t3.nano** instances:

ansible-playbook -i inventory_aws_ec2.yaml -l instance_type_t3_nano my-playbook.yaml

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-playbook -i inventory_aws_ec2.yaml -l instance_type_t3_nano my-playbook.yaml

PLAY [Deploy Nginx website and configuration files] **********************************

TASK [Gathering Facts] ***************************************************************
ok: [public_dns_name]

TASK [install php-fpm and php-curl] **************************************************
ok: [public_dns_name]

TASK [Copy website files] ************************************************************
ok: [public_dns_name]

TASK [Copy nginx.conf template] ******************************************************
ok: [public_dns_name]

TASK [Restart nginx] *****************************************************************
changed: [public_dns_name]

TASK [Start and enable php-fpm] ******************************************************
ok: [public_dns_name]

PLAY RECAP ***************************************************************************
public_dns_name            : ok=18   changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

9. **Update ansible.cfg** to use inventory by default:

inventory = ./inventory_aws_ec2.yaml

```
  GNU nano 7.2                              ansible.cfg *
[defaults]
inventory = ./inventory_aws_ec2.yaml
host_key_checking = False
interpreter_python = /usr/bin/python3
deprecation_warnings = False
enable_plugins = aws_ec2
private_key_file = ~/.ssh/id_ed25519
```

10. Now you can simply run:

ansible-playbook -l instance_type_t3_nano my-playbook.yaml

```
(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $ ansible-playbook -l instance_type_t3_nano my-playbook.yaml
[WARNING]: Ansible is being run in a world writable directory (/workspaces/terraform_machine), ignoring it as an ansible.cfg source. For more information see https://docs.ansible.com/ansible/devel/reference_appendices/config.html#cfg-in-world-writable-dir
[WARNING]: No inventory was parsed, only implicit localhost is available
[WARNING]: provided hosts list is empty, only localhost is available. Note that the implicit localhost does not match 'all'
[WARNING]: Could not match supplied host pattern, ignoring: instance_type_t3_nano

PLAY [Configure nginx web server] *******************************************************
skipping: no hosts matched

PLAY [Configure SSL certificates] *******************************************************
skipping: no hosts matched

PLAY [Deploy Nginx website and configuration files] *************************************
skipping: no hosts matched

PLAY RECAP *******************************************************************************

(ansible-core) @manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

Task 13 – Ansible roles: nginx, ssl, webapp

Reorganize your configuration into **roles**.

1. **Update main.tf** for a simple dev environment with 1 instance (as shown previously).

```
 GNU nano 7.2                              main.tf *
provider "aws" {
  shared_config_files      = ["~/.aws/config"]
  shared_credentials_files = ["~/.aws/credentials"]
}

resource "aws_vpc" "myapp_vpc" {
  cidr_block           = var.vpc_cidr_block
  enable_dns_hostnames = true
  tags = {
     Name = "${var.env_prefix}-vpc"
  }
}

module "myapp-subnet" {
  source                 = "./modules/subnet"
  vpc_id                 = aws_vpc.myapp_vpc.id
  subnet_cidr_block      = var.subnet_cidr_block
  availability_zone      = var.availability_zone
  env_prefix             = var.env_prefix
  default_route_table_id = aws_vpc.myapp_vpc.default_route_table_id
}

module "myapp-webserver" {
  source              = "./modules/webserver"

^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execute     ^C Location
```

2. **Create /ansible structure**:

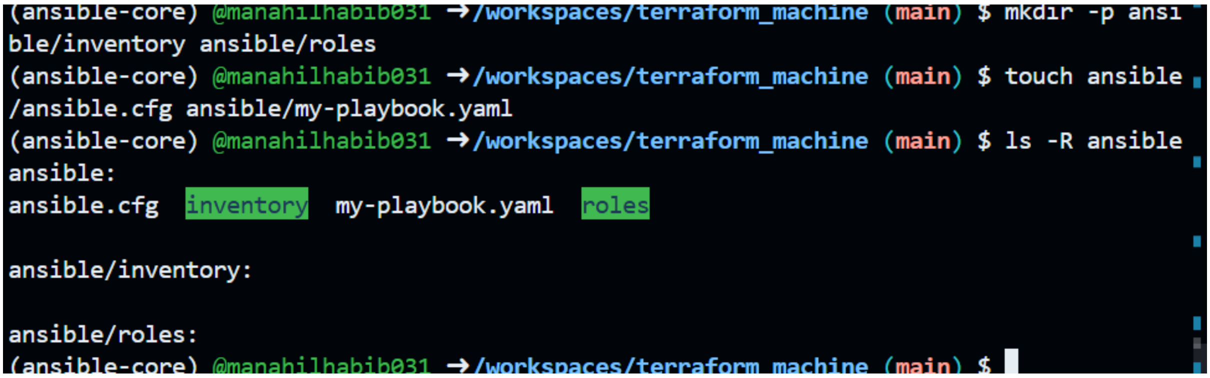
mkdir -p ansible

cd ansible

mkdir inventory roles

touch ansible.cfg my-playbook.yaml

ls -R

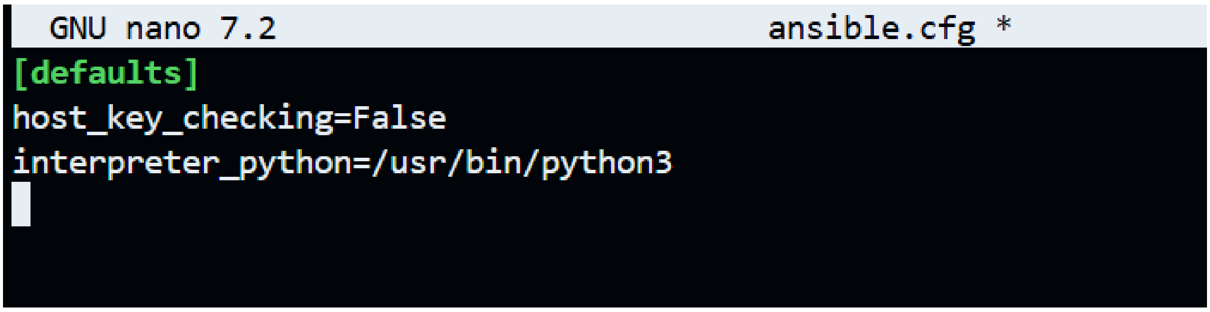


3. **ansible/ansible.cfg**:

[defaults]

host_key_checking=False

interpreter_python = /usr/bin/python3



4. **ansible/inventory/hosts**:

[nginx]

<public-ip-of-machine>

[nginx:vars]

ansible_ssh_private_key_file=~/.ssh/id_ed25519

ansible_user=ec2-user

```
  GNU nano 7.2                      ansible/inventory/hosts *
[nginx]
<public-ip-of-machine>

[nginx:vars]
ansible_ssh_private_key_file=~/.ssh/id_ed25519
ansible_user=ec2-user
```

5. **Create roles**:

cd roles

ansible-galaxy role init nginx

ansible-galaxy role init ssl

ansible-galaxy role init webapp

cd ..

ls -R

```
(ansible-core) @manahilhabib031 ➜ /workspaces/terraform_machine/ansible (main) $ ls -R
 roles

roles/webapp:
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars

roles/webapp/defaults:
main.yml

roles/webapp/files:

roles/webapp/handlers:
main.yml

roles/webapp/meta:
main.yml

roles/webapp/tasks:
main.yml

roles/webapp/templates:

roles/webapp/tests:
inventory  test.yml

roles/webapp/vars:
main.yml
(ansible-core) @manahilhabib031 ➜ /workspaces/terraform_machine/ansible (main) $
```

6. **Role: nginx**

- ansible/roles/nginx/handlers/main.yml:

```
GNU nano 7.2                ansible/roles/nginx/handlers/main.yml *
- name: Restart nginx
  service:
    name: nginx
    state: restarted
```

- ansible/roles/nginx/tasks/main.yml:

```
GNU nano 7.2                ansible/roles/nginx/handlers/main.yml *
- name: Install nginx
  yum:
    name: nginx
    state: present
    update_cache: yes
  notify: Restart nginx

- name: Install openssl
  yum:
    name: openssl
    state: present

- name: Start and enable nginx
  service:
    name: nginx
    state: started
    enabled: true
```

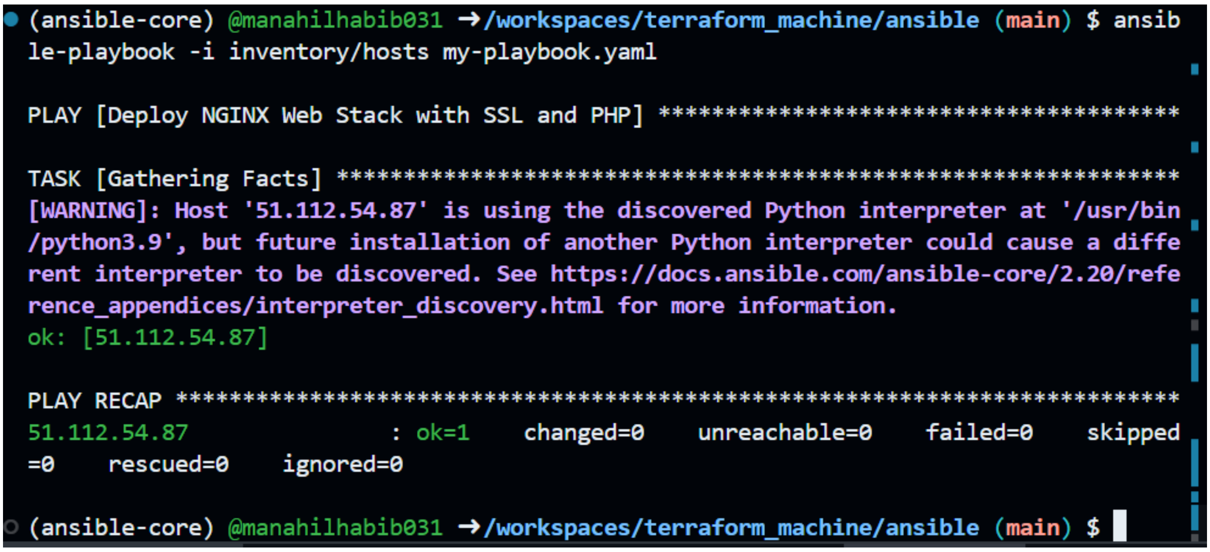
7. **First role-based playbook** ansible/my-playbook.yaml:

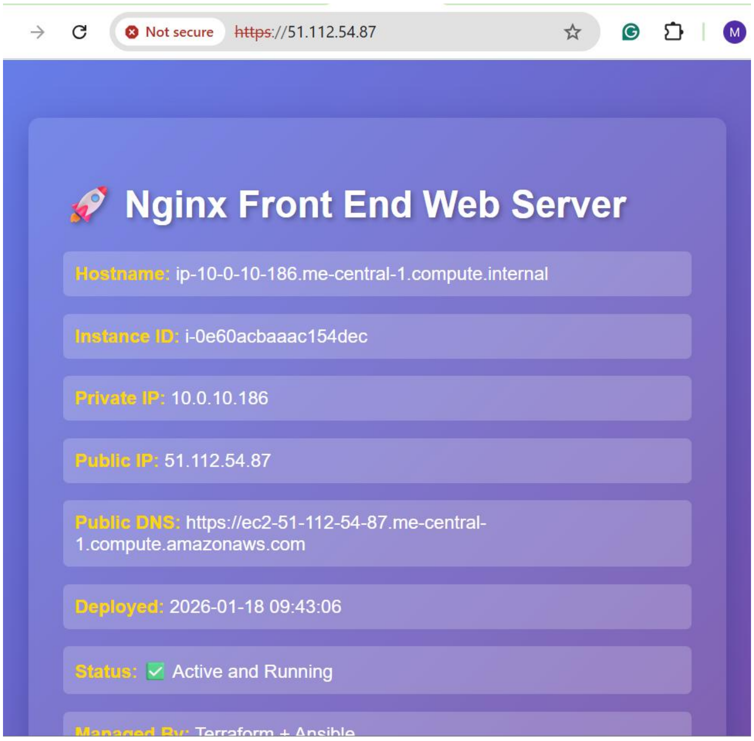
```
GNU nano 7.2                              my-playbook.yaml *
---
- name: Deploy NGINX Web Stack with SSL and PHP
  hosts: nginx
  become: true
  roles:
    - nginx
```

Run:

chmod 755 $(pwd)

ansible-playbook -i inventory/hosts my-playbook.yaml

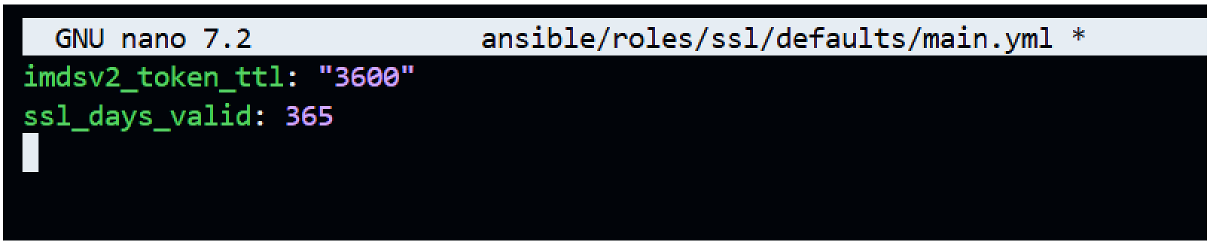




- **Role: ssl**
- ansible/roles/ssl/defaults/main.yml:

imdsv2_token_ttl: "3600"

ssl_days_valid: 365



- ansible/roles/ssl/tasks/main.yml:

```
  GNU nano 7.2                 ansible/roles/ssl/tasks/main.yml *
- name: Create SSL private directory
  file:
    path: /etc/ssl/private
    state: directory
    mode: '0700'

- name: Create SSL certs directory
  file:
    path: /etc/ssl/certs
    state: directory
    mode: '0755'

- name: Get IMDSv2 token
  uri:
    url: http://169.254.169.254/latest/api/token
    method: PUT
    headers:
      X-aws-ec2-metadata-token-ttl-seconds: "{{ imdsv2_token_ttl }}"
    return_content: yes
  register: imds_token

- name: Get public IP
  uri:
    url: http://169.254.169.254/latest/meta-data/public-ipv4

^G Help          ^O Write Out     ^W Where Is      ^K Cut           ^T Execute       ^C Location
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line
```

9. **Role: webapp**

- ansible/roles/webapp/defaults/main.yml:

```
  GNU nano 7.2              ansible/roles/webapp/defaults/main.yml *
nginx_user: nginx
nginx_worker_processes: auto
nginx_worker_connections: 1024
nginx_error_log_level: notice

web_root: /usr/share/nginx/html
web_index_file: index.php
```

- ansible/roles/webapp/files/index.php – PHP metadata page.

```
  GNU nano 7.2                ansible/roles/webapp/files/index.php *
<?php
// Get hostname
$hostname = gethostname();

// Deployment date
$deployed_date = date("Y-m-d H:i:s");

// Metadata base URL
$metadata_base = "http://169.254.169.254/latest/";

// Function to get IMDSv2 token
function getImdsV2Token() {
    $ch = curl_init("http://169.254.169.254/latest/api/token");
    curl_setopt_array($ch, [
        CURLOPT_RETURNTRANSFER => true,
        CURLOPT_CUSTOMREQUEST  => "PUT",
        CURLOPT_HTTPHEADER     => [
            "X-aws-ec2-metadata-token-ttl-seconds: 21600"
        ],
        CURLOPT_TIMEOUT        => 2
    ]);

    $token = curl_exec($ch);
    curl_close($ch);
```

- ansible/roles/webapp/handlers/main.yml:

```
  GNU nano 7.2                ansible/roles/webapp/handlers/main.yml *
- name: Restart nginx
  service:
    name: nginx
    state: restarted

- name: Restart php-fpm
  service:
    name: php-fpm
    state: restarted
```

- ansible/roles/webapp/templates/nginx.conf.j2:

```
GNU nano 7.2          ansible/roles/webapp/templates/nginx.conf.j2 *
user {{ nginx_user }};
worker_processes {{ nginx_worker_processes }};
error_log /var/log/nginx/error.log {{ nginx_error_log_level }};
pid /run/nginx.pid;

events {
    worker_connections {{ nginx_worker_connections }};
}

http {
    log_format  main  '$remote_addr - $remote_user [$time_local] "$request"'
                      '$status $body_bytes_sent "$http_referer"'
                      '"$http_user_agent" "$http_x_forwarded_for"';

    access_log  /var/log/nginx/access.log  main;

    sendfile            on;
    tcp_nopush          on;
    keepalive_timeout   65;
    types_hash_max_size 4096;

    include             /etc/nginx/mime.types;
    default_type        application/octet-stream;

^G Help     ^O Write Out   ^W Where Is   ^K Cut     ^T Execute    ^C Location
```

- ansible/roles/webapp/tasks/main.yml:

```
GNU nano 7.2                 ansible/roles/webapp/tasks/main.yml *

- name: Deploy nginx config
  template:
    src: nginx.conf.j2
    dest: /etc/nginx/nginx.conf
  notify: Restart nginx

- name: Start and enable php-fpm
  service:
    name: php-fpm
    state: started
    enabled: true
```

10. **Final role-based playbook** ansible/my-playbook.yaml:

- name: Deploy NGINX Web Stack with SSL and PHP hosts: nginx

become: true

roles:

- nginx
- ssl
- webapp

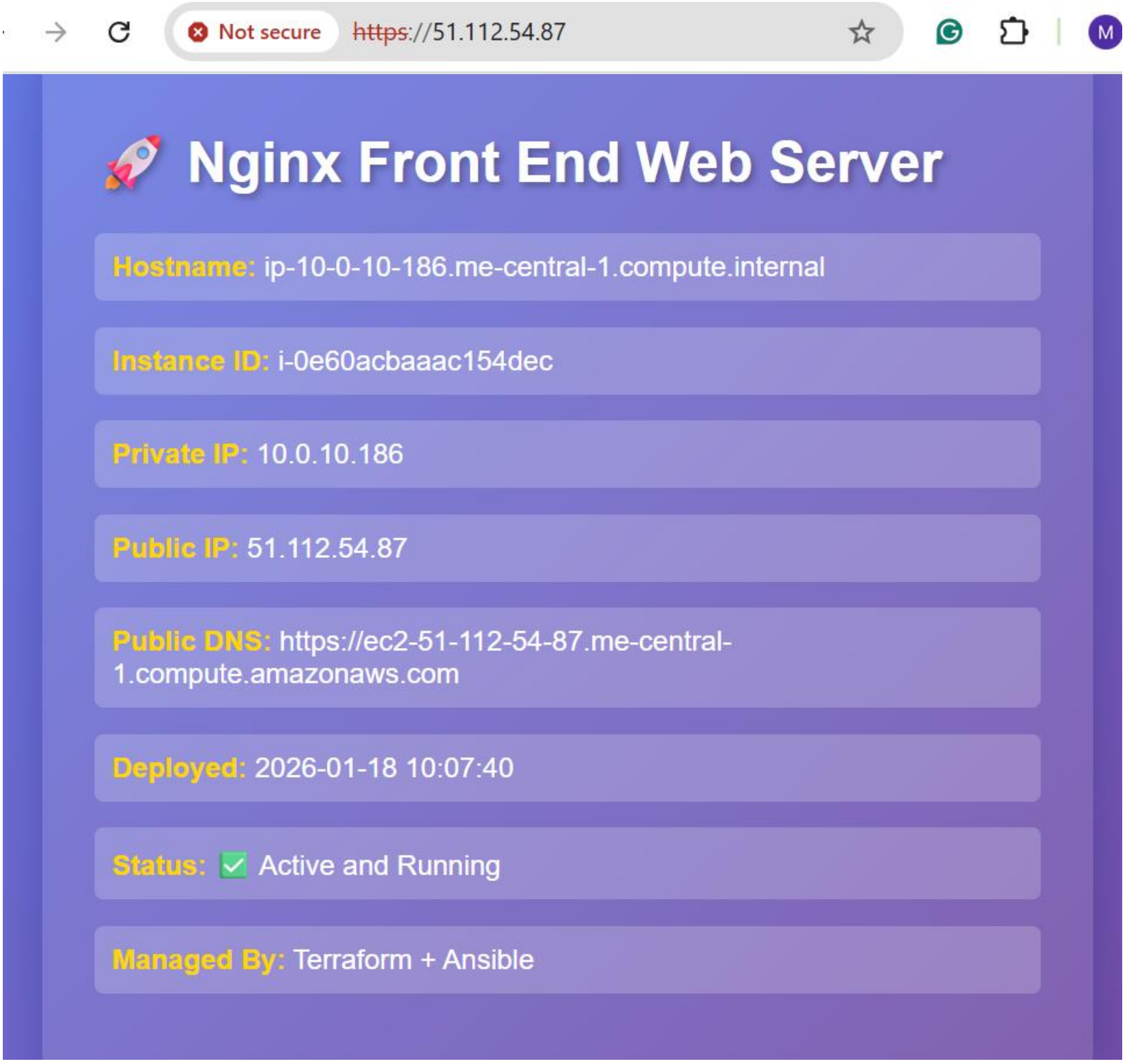
```
  GNU nano 7.2                    ansible/my-playbook.yaml *
---
- name: Deploy NGINX Web Stack with SSL and PHP
  hosts: nginx
  become: true
  roles:
    - nginx
    - ssl
    - webapp
```

11. Run:

ansible-playbook -i inventory/hosts my-playbook.yaml

```
(ansible-core) @manahilhabib031 →/workspaces/terraform_machine/ansible (main) $ ansib
le-playbook -i inventory/hosts my-playbook.yaml

PLAY [Deploy NGINX Web Stack with SSL and PHP] ***************************************

TASK [Gathering Facts] ***************************************************************
[WARNING]: Host '51.112.54.87' is using the discovered Python interpreter at '/usr/bin
/python3.9', but future installation of another Python interpreter could cause a diffe
rent interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/refe
rence_appendices/interpreter_discovery.html for more information.
ok: [51.112.54.87]

PLAY RECAP ***************************************************************************
51.112.54.87               : ok=1    changed=0    unreachable=0    failed=0    skipped
=0    rescued=0    ignored=0

(ansible-core) @manahilhabib031 →/workspaces/terraform_machine/ansible (main) $
```

Visit https://<public-ip> and verify the PHP page with metadata.



Cleanup

1. From the Terraform root:

terraform destroy -auto-approve

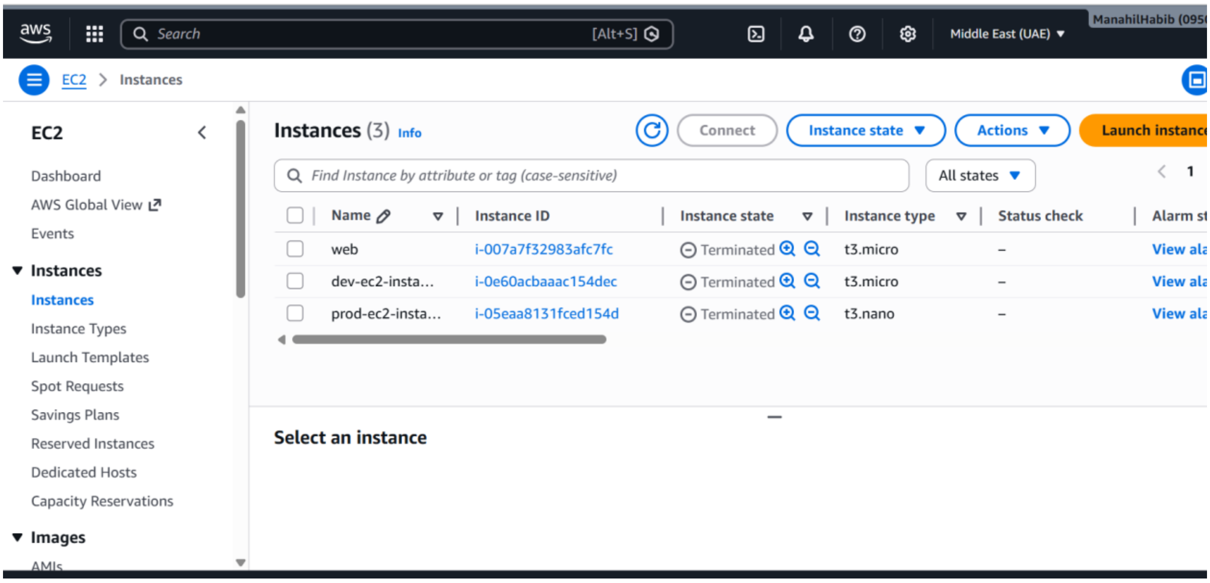
```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ terraform destroy -auto-approve

Changes to Outputs:
  - prod-webserver_public_ips = [
      - "158.252.83.205",
    ] -> null
  - webserver_public_ips      = [
      - "51.112.54.87",
    ] -> null
module.myapp-webserver[0].aws_key_pair.ssh_key: Destroying... [id=dev-serverkey-0]
module.myapp-subnet.aws_default_route_table.main_rt: Destroying... [id=rtb-0d0401b4a4956c429]
module.myapp-webserver-prod[0].aws_key_pair.ssh_key: Destroying... [id=prod-serverkey-0]
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destroying... [id=subnet-0d6d5c9bc692ce8b3]
module.myapp-webserver[0].aws_security_group.web_sg: Destroying... [id=sg-0f167917634e66dea]
module.myapp-subnet.aws_default_route_table.main_rt: Destruction complete after 0s
module.myapp-webserver-prod[0].aws_security_group.web_sg: Destroying... [id=sg-0f92a1341c890c281]
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destroying... [id=igw-0cb93b6c570723e31]
module.myapp-webserver[0].aws_key_pair.ssh_key: Destruction complete after 1s
module.myapp-webserver-prod[0].aws_key_pair.ssh_key: Destruction complete after 1s
module.myapp-subnet.aws_subnet.myapp_subnet_1: Destruction complete after 1s
module.myapp-subnet.aws_internet_gateway.myapp_igw: Destruction complete after 1s
module.myapp-webserver-prod[0].aws_security_group.web_sg: Destruction complete after 1s
module.myapp-webserver[0].aws_security_group.web_sg: Destruction complete after 1s
aws_vpc.myapp_vpc: Destroying... [id=vpc-0cf8ec53f8167c1f7]
aws_vpc.myapp_vpc: Destruction complete after 1s

Destroy complete! Resources: 8 destroyed.
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
paces: legendary lamp   main*   ⊗ 0 ⚠ 0   0
```

2. Verify state:

cat terraform.tfstate

```
@manahilhabib031 ➜/workspaces/terraform_machine (main) $ cat terraform.tfstate
{
  "version": 4,
  "terraform_version": "1.14.3",
  "serial": 92,
  "lineage": "babe551c-4973-02bc-ff3c-a91c60a01112",
  "outputs": {},
  "resources": [],
  "check_results": null
}
@manahilhabib031 ➜/workspaces/terraform_machine (main) $
```

3. Confirm that no EC2 instances remain in AWS console.


aws
Search
[Alt+S]
Middle East (UAE)
ManahilHabib (095
EC2
Instances
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
Instances (3) Info
Connect
Instance state
Actions
Launch instance
Find Instance by attribute or tag (case-sensitive)
All states
1
Name
Instance ID
Instance state
Instance type
Status check
Alarm st
web
i-007a7f32983afc7fc
Terminated
t3.micro
–
View ala
dev-ec2-insta...
i-0e60acbaaac154dec
Terminated
t3.micro
–
View ala
prod-ec2-insta...
i-05eaa8131fced154d
Terminated
t3.nano
–
View ala
Select an instance

****